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What influences algorithmic decision-making? A systematic literature review on algorithm aversion

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ABSTRACT

With the continuing application of artificial intelligence (AI) technologies in decision-making, algorithmic decision-making is becoming more efficient, often even outperforming humans. Despite this superior performance, people often consciously or unconsciously display reluctance to rely on algorithms, a phenomenon known as algorithm aversion. Viewed as a behavioral anomaly, algorithm aversion has recently attracted much scholarly attention. With a view to synthesize the findings of existing literature, we systematically review 80 empirical studies identified through searching in seven academic databases and using the snowballing technique. We inductively categorize the influencing factors of algorithm aversion under four main themes: algorithm, individual, task, and high-level. Our analysis reveals that although algorithm and individual factors have been investigated extensively, very little attention has been given to exploring the task and high-level factors. We contribute to algorithm aversion literature by proposing a comprehensive framework, highlighting open issues in existing studies, and outlining several research avenues that could be handled in future research. Our model could guide developers in designing and developing and managers in implementing and using of algorithmic decision.

1. Introduction

Increased sophistication of digital technology, rapidly flourishing big data analytics, and continuing development of artificial intelligence (AI) technologies such as machine learning (ML) and natural language processing enable individuals and organizations to rely on algorithmic decisions. Today, the application of algorithms in decision-making can be seen, to some extent, in a wide range of areas, spanning from objective tasks to tasks that require subjective evaluation. For example, in healthcare, algorithms are used for imaging, diagnosis, surgery, and personalizing medicine (Castelluccia and Le Métayer, 2019). Robo-advisors are utilized to make investment decisions (Zhang et al., 2021). Managers use algorithms to forecast the future demand of products (Kawaguchi, 2021) and hire employees (Diab et al., 2011). Marketers use algorithms for research, strategy formulation, channel management, and performance evaluation (De Bruyn et al., 2020; Huang and Rust, 2020; Vlačić et al., 2021). Chatbots take the role of customer service (Rapp et al., 2021). However, people's reactions to algorithms are not straightforward. Studies show that these reactions are often influenced by various factors (Berger et al., 2021; Dietvorst et al., 2015; Kawaguchi, 2021). While some people prefer algorithms, some do not (McBride et al., 2012; Sharan and Romano, 2020). While for some tasks, algorithms are highly sought after, in others they might evoke rejection (Bogert et al., 2021; Prahl and Swol, 2021). More intriguingly, people reject algorithms despite being familiar with their superior performance, a tendency known as algorithm aversion (Dietvorst et al., 2015). Algorithm aversion can be viewed as a behavior of neglecting algorithmic decisions in favor of one's own decisions or other's decisions, either consciously or unconsciously. It is logical to follow algorithms when it is evident that algorithms perform better than humans. Deviating from this rational behavior may be viewed as a behavioral anomaly, which may reduce human subjects' expected utility (Dietvorst et al., 2015; Filiz et al., 2021). In a situation (e.g., medical diagnosis, bail or sentence decision) when a bad decision may lead to serious consequences, such an anomaly may have a significant impact. Therefore, understanding the drivers of algorithm aversion is important to leverage the benefits of algorithmic decisions to the fullest.

Recently, researchers have demonstrated an increasing interest in

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the study of algorithm aversion. Numerous studies have been conducted to investigate the factors responsible for aversion and to suggest solutions thereof. Notwithstanding the proliferation of such studies, we have limited scope to obtain a comprehensive overview regarding algorithm aversion since extant literature, individually, tends to focus on only a limited number of elements of algorithm aversion. A review-based study, which summarizes existing studies, can assist in this regard (Kitchenham and Charters, 2007; Tandon et al., 2020). Thus, we conduct a systematic literature review (SLR) to synthesize the existing body of knowledge on algorithm aversion and to provide a comprehensive framework to direct future studies (Kitchenham and Charters, 2007). Our SLR is guided by the following research questions (RQs): **RQ1.** What factors influence algorithm aversion? **RQ2.** What are the research gaps and limitations of the prior literature and what future research opportunities can be derived to advance the use of algorithmic decisions?

To address these specific research questions, the current study involves a review of 80 studies identified through searching in seven academic databases and partaking in extensive citation chaining. We report the current state of the research through descriptive analysis and derive themes following an inductive approach (Braun and Clarke, 2006). Our thematic analysis reflects a wide range of factors responsible for algorithm aversion, which we classify into four main themes: algorithm, individual, task, and high-level. We highlight the gaps and limitations of previous research and utilize these gaps to suggest future research directions. Additionally, a comprehensive framework has been presented, delineating the relationships of different factors emerging from this study to provide a holistic overview of algorithm aversion.

Our study contributes both theoretically and practically to algorithm aversion literature (Berger et al., 2021; Dietvorst et al., 2015; Prahl and van Swol, 2017). First, we find that most studies are scenario-based online or laboratory experiments. Hence, we highlight the need for research in real-world settings. Second, current study provides a synthesized framework of algorithm aversion, which can serve as a groundwork for future scholars. Third, we underscore the importance of developing a measurement scale of algorithm aversion to test and validate the existing findings as well as to conduct new applied research in real-world settings. Fourth, we highlight the significance of studying organizational factors, which have been neglected in algorithm aversion research. Fifth, our research proposed several future research directions with special emphasis on AI algorithmic decision, which draws relatively less attention from the research community. Sixth, the present study provides important guidance for algorithm developers to design algorithms more effectively. Seventh, managers may find the results of this study useful while using algorithmic decisions in their organizations.

The remainder of the study is organized as follows. Section 2 provides a brief description of algorithms and algorithm aversion. Section 3 explicates the research methods. Section 4 explains the findings of the current study. Section 5 depicts a comprehensive framework of algorithm aversion. Section 6 highlights the research gaps and future research directions. Section 7 outlines general discussion and implications of the study. Finally, section 8 concludes the study.

2. Algorithms and algorithm aversion

An algorithm is a "set of encoded procedures for transforming input data into a desired output, based on specified calculations" (Gillespie, 2014, p. 1). Echoing Araujo et al. (2020), this study assumes an algorithm is an automated process that provides decisions independently without the mediation of a human. Algorithms use data, statistics, and/or computing resources to make decisions. Algorithms used to rely mostly on statistical methods such as regression analysis to make decisions. However, recently, AI, which refers to "the ability of a system to identify, interpret, make inferences, and learn from data to achieve predetermined organizational and societal goals" (Mikalef and Gupta, 2021), has been incorporated into algorithms to make them more powerful. This development enables algorithms to learn independently over time from data to discover patterns in the data and make better decisions. While in systems without AI, humans decide which factors are to be considered by algorithms for decision-making, in systems with AI, algorithms are free to select factors based on the patterns discovered. In some contexts (e.g., product recommendations by Amazon, self-driving cars, and automated metro systems), algorithms can be used to completely replace human decisions. However, in most real-world cases, algorithms play a supporting role in decision-making by humans (Acharya et al., 2018). For example, algorithms can assist bankers in identifying eligible customers for credit and healthcare practitioners in diagnosing diseases.

In algorithm aversion literature, various terms have been used to refer to algorithmic decision-making (Burton et al., 2020; Jussupow et al., 2020). In our current study, we consider "algorithm" as an umbrella term that conveys the related notions such as AI-generated decision (Köbis and Mossink, 2021), robo-advisors (Niszczota and Kaszás, 2020; Zhang et al., 2021), digital agent (Stein et al., 2020), decision support system (Feng et al., 2018; Lim and O'Connor, 1996; Shaffer et al., 2012), automated (automation) decision/advisor/aid (Araujo et al., 2020; Dzindolet et al., 2002; Pearson et al., 2019), chatbot interaction (Luo et al., 2019), machine decision/agents (Bigman and Gray, 2018; Goodyear et al., 2017), mechanical decision (Lee, 2018), forecasting support system (Goodwin et al., 2013), actuarial decision (Eastwood et al., 2012), statistical advice (Önkal et al., 2009), robot recommendations (Araujo et al., 2020; Qiu and Benbasat, 2008; Rau et al., 2009), computer decision (Promberger and Baron, 2006), medication management system (Ho et al., 2005), and expert system (Dijkstra, 1999; Dijkstra et al., 1998; Swinney, 1999; Workman, 2005).

Research found that algorithms consistently outperform humans in decision-making (Dietvorst et al., 2015; Elkins et al., 2014). Despite this superior performance, people tend to reject algorithms and choose human decision-makers (Dietvorst et al., 2015). Algorithm aversion is also exhibited even when the accuracy of the decisions of both algorithms and humans are identical (Berger et al., 2021; Bogert et al., 2021). In addition to displaying aversion while comparing algorithmic decisions against human experts' decisions, people also show aversion when they compare their own decisions against algorithmic decisions (Kawaguchi, 2021; Litterscheidt and Streich, 2020; Stein et al., 2020; Sultana et al., 2021). Scholars have also discovered that some people are intrinsically averse to algorithms, irrespective of their performance, due to their fundamental distrust of algorithms (Kawaguchi, 2021; Prahl and van Swol, 2017). As such, we define algorithm aversion as *a behavior of discounting algorithmic decisions with respect to one's own decisions or other's decisions, either consciously or unconsciously*. Our definition regards algorithm aversion as a behavior, as it is an observable action (BSSR, 2019; Filiz et al., 2021). We argue that people discount algorithmic decisions "consciously or unconsciously" as we see some people reject algorithmic decisions upon deliberately thinking about the performance of these decisions while others discount them simply by habit without considering their performance.

3. Research method

To analyze prior research, the current study adopts a systematic literature review, which involves "identifying, evaluating and interpreting all available research relevant to a particular research question, or topic area, or phenomenon of interest" (Kitchenham and Charters, 2007). SLR provides "robust, reliable summaries of the most reliable evidence" (Petticrew and Roberts, 2008). To perform SLR, we follow an inductive process to derive themes of influencing factors of algorithm aversion. In information system research, many SLRs have been conducted using inductive methods to derive relevant themes (e.g., Banijamali et al., 2020; Behutiye et al., 2017; Bhimani et al., 2019; Dikert et al., 2016; Guckenbiehl et al., 2021; Taušan et al., 2017).

To ensure the reliability, transferability, and transparency of our study, we also adopt the process suggested by Kitchenham and Charters (2007), with a particular emphasis on documenting the search strategy. The SLR method proposed by Kitchenham and Charters (2007) has also been used extensively in information system research (e.g., Bano and Zowghi, 2015; Garousi et al., 2016; Genc-Nayebi and Abran, 2017; Qazi et al., 2015). As such, considering the above discussion, we follow six steps in our review (Fig. 1): i) Define: specifying inclusion, exclusion, and quality assessment criteria of the study; ii) Search: deciding on search terms and databases and conducting queries on databases; iii) Select: selecting pertinent studies through filtering processes and using the snowballing technique to optimize the selection; iv) Extract and synthesize: systematically extract data and synthesize for analysis; v) Analyze: analyze the data descriptively to derive themes. Finally, findings are presented with analytical discussion.

3.1. *Defining study selection criteria*

To base our review on quality evidence, we define inclusion, exclusion, and quality assessment criteria, considering the scope of the review (Tranfield et al., 2003). Only the studies that strictly meet all the criteria are considered for review.

3.1.1. *Inclusion criteria*

We adopt the following criteria to select a particular study to be included in the review process.

i) The study has to discuss algorithmic decision making. This criterion allows us to exclude studies that discuss only algorithms, algorithms' performance (Barbosa et al., 2020), the effects of algorithms (Cheyne et al., 2008; Hu, 1992), or perception towards algorithms disregarding the decision component (Kolbinger and Knopp, 2020).
ii) The study has to deal with interaction between humans and algorithms while making decisions. This criterion is set to exclude studies that discuss only algorithmic decision, human decision making, or comparison between different algorithmic decisions (Sakthivel and Sathya, 2021; Yazdani-Asrami et al., 2020) or between different humans' decisions.
iii) We consider studies that were published any time before March 2021.
iv) The study has to have been published in a peer-reviewed journal and be written in English.
v) The study must be based on empirical research. Since human-algorithm research is largely based on observation and experience (MacKenzie, 2012), our focus on empirical research is quite representative.
vi) Full text of the study has to be available online.

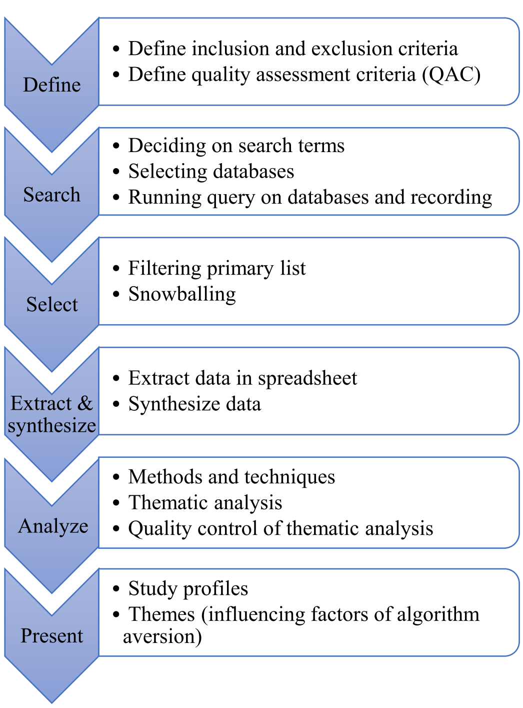

Fig. 1. Review process.

3.1.2. *Exclusion criteria*

Further to the above inclusion criteria, we follow the following exclusion criteria to have a more representative corpus of studies.

i) Conference paper, book chapters, book, position papers, review papers, short papers (i.e., less than 6 pages (Zahedi et al., 2016)), workshop papers, theses, panel discussions, posters, editorials, commentary, keynotes, tutorial summaries, and technical papers are excluded.
ii) We exclude all those studies that do not discuss algorithm aversion, appreciation, or adoption in decision-making context(s).
iii) We eliminate all those studies that do not satisfy the minimum threshold limit of quality assessment criteria.
iv) We do not consider study that is a literature review on the current topic.

3.1.3. *Quality assessment criteria*

In addition to the above inclusion and exclusion criteria, we apply separate subjective quality assessment criteria (QAC) to filter out studies irrelevant to our study. The QAC are comprised of five items involving alignment of the study with the objectives of the current study, description of research method, whether the study reports aversion or appreciation, and the credibility of the findings (Table 1). We adapt the items from prior literature review studies and set scoring criteria based on our subjective judgment as well as guided by prior works (Behera et al., 2019; Khan et al., 2021; Tandon et al., 2020). The highest achievable score is 12. For the current study, we set benchmark at a score of 6, that is, 50%, for qualification, following previous studies (Behera et al., 2019; Khan et al., 2021; Tandon et al., 2020).

3.2. *Search strategy*

Our search strategy is composed of three sequential steps: deciding on specific search terms, selecting databases, and running search queries on selected databases.

3.2.1. *Search terms*

Inspired by the seminal work of Dietvorst et al. (2015), we initially used two search terms, "algorithm aversion" and "algorithm

Table 1
Quality Assessment Criteria (QAC).

Sl. No.	Criteria	Score
QAC1	Study objectives are clearly stated and in line with this study	Yes (+2), partially (+1), No (+0)
QAC2	Research method is described adequately (research method, data collection, analysis, etc.)	Yes (+2), partially (+1), No (+0)
QAC3	Study discusses the reasons for algorithm aversion/appreciation	Yes (+3), partially (+1.5), No (+0)
QAC4	Study reports the method of overcoming algorithm aversion/increasing appreciation	Yes (+3), partially (+1.5), No (+0)
QAC5	Findings of the study are plausible (Behera et al., 2019)	Yes (+2), partially (+1), No (+0)

Table 2
Search terms.

ID	Search syntax	Total hits	Ultimately retained**
	Themes of search term: algorithm, artificial intelligence, and machine learning; decision, advice, recommendation, and decision aid		
1	Algorithm* Aversion	840	16
2	Algorithm* Appreciation	788	3
3	(AI OR "Artificial Intelligence") AND Aversion	162	2
4	(AI OR "Artificial Intelligence") AND Appreciation	197	–
5	"AI recommendation" OR "Artificial intelligence recommendation" OR "Algorithm* recommendations" OR "Machine Learning recommendation" OR "ML recommendation"	249	1
6	"AI decision*" OR "Artificial intelligence decision*" OR "Algorithm* decision*" OR "Machine Learning decision*" OR "ML decision*"	2009	3
7	"AI Advice" OR "Artificial intelligence Advice" OR "Algorithm* advice" OR "Machine Learning advice" OR "ML advice"	15	3
8	("AI" OR "Artificial intelligence" OR "Algorithm*" OR "Machine Learning" OR "ML") AND "Decision aid"	697	5

[*] Takes the place of one or more characters in the search term.
[**] Figures represent the number of studies after completing selection process.

appreciation" on a pilot basis in "Scopus" and "ACM DL (Full-text)" databases. The search resulted in 22 studies in Scopus and 4 studies in ACM DL with no duplication. We read those studies and identified other relevant search terms. Since our focus was on algorithm aversion, which are outcomes of decision interaction, we decided to include the term "decision" and its variants such as advice, recommendations, and decision aid in our search terms list. Furthermore, the literature revealed that several other terms such as artificial intelligence and machine learning were used in exchange for algorithms in the context of algorithmic decision. Such a revelation motivated us to integrate these terms along with their abbreviated forms. During search term selection, all authors regularly reviewed the process through several iterations and refinement, following the practice of other research reviews (Rapp et al., 2021; ter Stal et al., 2020). The final list of search terms can be found in Table 2.

3.2.2. Selecting databases

Going through the studies found in the pilot search, it was revealed to us that the topic at hand encompasses a wide range of disciplines such as information systems, psychology, healthcare, and business. Contemplating this finding, we decided to select multiple databases to capture potentially any study that addresses the topic at hand. We chose seven databases, namely Scopus, Web of Science, the Institute of Electrical and Electronics Engineers (IEEE), the Association for Computing Machinery Digital Library (ACM DL), EBSCO*host* (utilizing both Academic Search Elite and Business Source Complete), PubMed, and JSTOR. Scopus, Web of Science, EBSCO*host*, and JSTOR were chosen due to their extensive coverages of multiple disciplines, notably life science, social science, humanities, business, psychology, and health science (Kaur et al., 2021). ACM, DL, and IEEE were selected as they are relevant sources of studies related to human-computer interactions (Rapp et al., 2021). Finally, PubMed was chosen for its special foci on biomedical science and interdisciplinary research (Behera et al., 2019). Upon running search stings in these databases, we found a copious number of duplicate studies (2031) across databases, which indicates the adequacy of the

Table 3
Database.

ID	Database	URL	Total hits	Duplicate	Ultimately retained*
1	Scopus	http://www.scopus.com/home.url	1886	46	27
2	WoS	http://www.webofknowledge.com	1473	1147	3
3	IEEE	https://ieeexplore.ieee.org	644	103	–
4	EBSCO*host*	https://search.ebscohost.com	483	397	2
5	ACM DL (Full-text)	http://dl.acm.org	19	9	1
6	PubMed	https://pubmed.ncbi.nlm.nih.gov/	429	317	–
7	JSTOR	https://www.jstor.org/	23	12	–

[*] Figures represent the number of studies after completing the selection process.

number of databases (Table 3).

3.2.3. Search process

In this stage, we ran search syntax in the selected databases. Search was restricted to journal study, study title, abstract, and keyword. We limited our search to studies that are published in the English language. There was no restriction on timeframe. We used both Boolean operators (AND, OR) and approximate phrase (wildcards, braces, quotation marks) for a more comprehensive search result (Collins et al., 2021). The search results were then exported to CSV, Excel, or any other suitable format supported by databases. Exported files that were not in Excel format were converted into Excel format for the sake of simplicity of future analysis. All studies were then assigned serial numbers along with search term ID and database ID so that studies could be tracked across the analytical steps. In the final stage, we merged all files into a single Excel file, rearranging the column headings into a unique sequence.

3.3. Study selection strategy

Our databases search primarily resulted in a total of 4957 studies. At this stage, the primary search result was refined through four linear filtering processes: i) redundancy check, ii) relevancy of title and abstract, iii) availability of full text, and iv) quality of the full text in relation to the current study. The study selection process and the number of removed and retained studies in each stage are depicted in Fig. 2.

In the first stage, we removed duplicate studies by matching Digital Object Identifier (DOI) where available and then title of the studies. After removing duplicate studies, we retained 2926 studies. Then, preliminary screening was performed through reading the title and abstract of the studies. While screening, studies were grouped into three categories: retain, remove, and suspect. There were 2733 removed (including 1 SLR study) and 89 retained studies. The high number of removed studies could partly be attributed to the bulk number of studies generated from financial "risk" domains such as studies about financial risk aversion using algorithms. There were 104 suspect studies whose admittance decisions were ironed out upon full-text reading and discussion among the authors. Among 193 studies (89 retained and 104 suspect), we could not download 12 studies (4 from retained and 8 from suspect), which resulted in 181 studies for next level filtering.

We examined the contents of suspect studies in light of the inclusion/exclusion criteria and quality assessment checklist. The examination

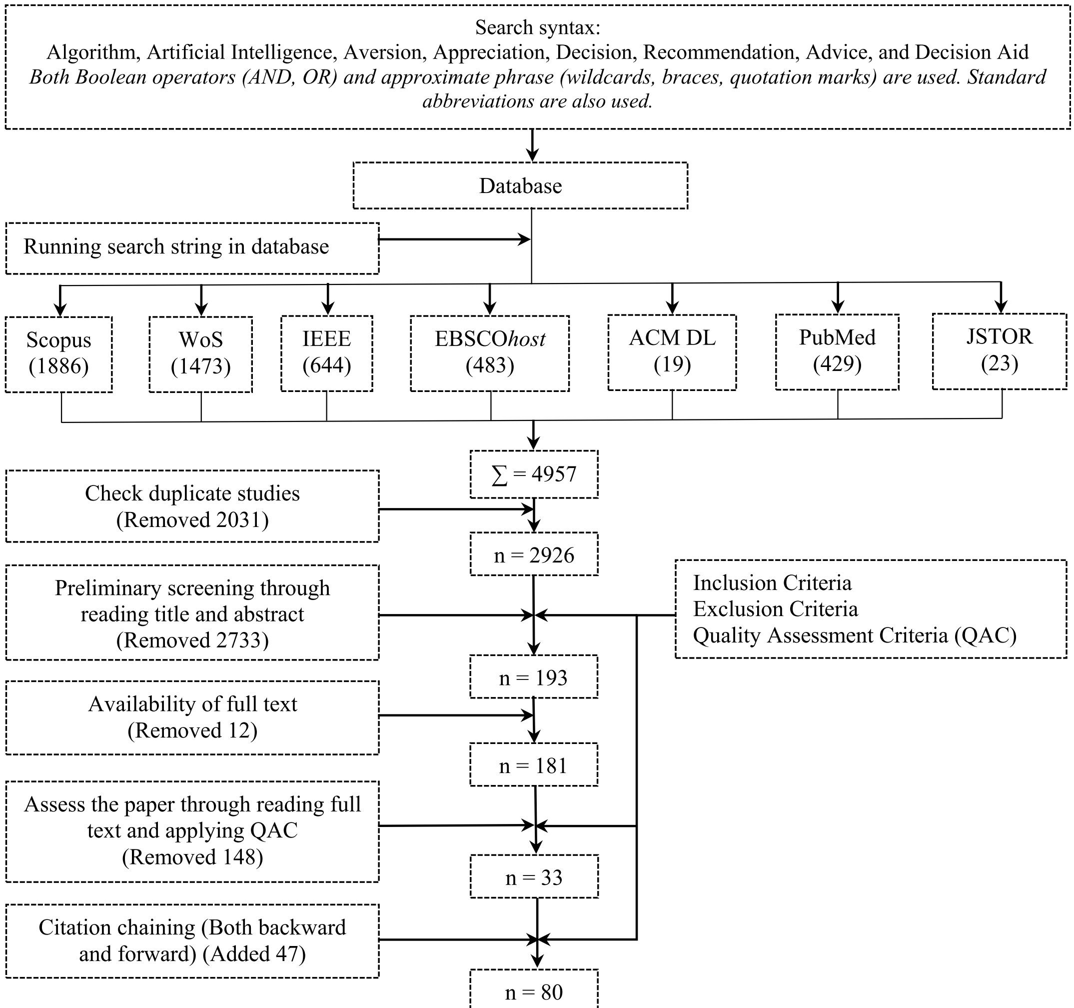

Fig. 2. Study selection process.

revealed that none of the suspect studies was qualified for further study. Then, the remaining studies (retain studies) were evaluated independently by the authors following the mentioned criteria. In case of any doubt, decision was reached by consensus; this occurred in 27 cases. This process guarantees a high degree of internal reliability (Minola et al., 2014). At this stage, we were able to retain 33 studies. Finally, we applied the snowballing technique (Wohlin, 2014), commonly known as citation chaining, for the primary studies published on or after 2015, the year in which the most influential work of Dietvorst et al. (2015) was published. Surprisingly, this step yielded 47 additional studies (37 from backward citation and 10 from forward citation). The ingress of such a high number of studies can be explained by the use of varied terms to convey the similar meaning of algorithmic decisions, as we discussed before. Since accommodating such a magnitude of variation is not viable in a database search, it is not surprising to miss some relevant studies in database search and capture those through the snowballing technique. Finally, our sample consisted of 80 studies, the quality scores (QS) of which are presented in Appendix A.

3.4. Data extraction and synthesis

The data extraction phase of a SLR study pertains to identifying and extracting pertinent information from each of the selected studies to seek answers to the research questions (Zahedi et al., 2016). First, the selected studies are arranged in chronological way. Then, first and second authors extract data in Excel sheets upon reading the studies against predefined items. After extracting data from every 20 studies, extracted data of each author are compared and discrepancies are resolved through discussion. These efforts ensure the consistency of data extraction (Jones et al., 2011). Appendix B shows the key items for data extraction, description of the items and their relevancy to the current study, and category of the data.

Extracted data were synthesized into three broad categories: protocol data, contextual/demographic data, and data for thematic analysis. Protocol data included data that were used in evaluating the permissibility of the study for review, for example, data related to inclusion/exclusion criteria and QAC. Contextual and demographic data are collected to understand the research profile such as research method used, type of data collection, and country of study. Finally, data related to key findings of the studies and influencing factors of algorithm aversion were extracted to derive themes. After extraction, data related to “influencing factors of algorithm aversion” and key “findings” of the studies are transferred into separate Excel sheets for further analysis.

3.5. Data analysis

To conduct thematic analysis, we followed the following steps. First, we read the Excel data file related to key findings of primary studies, and then, based on our understandings from the reading and experience in general, we assigned a descriptive statement to each study, outlining the focus of the study. This step enabled us to familiarize ourselves with the study more thoroughly. Second, since we were interested in exploring

Table 4
Supertheme: Algorithm; sub-themes and theme description.

T4	T3	T2	T1	Theme description
Design	Black box nature	Interactivity	Accessibility	Accessibility of people to algorithms' rationale.
			Explainability	Algorithms' ability to provide explanations with decisions.
			Understandability	Understandability of algorithms' explanations to users.
			Calibration feedback	Feedback is provided iteratively to calibrate trust.
			Feedback only	Feedback is provided without iterative process.
			Integration	Integrating people's opinion in algorithmic decision-making process.
			Algorithm's complexity	Complexity of algorithm by design.
			Response speed	The time an algorithm takes to provide decisions.
			Ability to learn	The ability of algorithms to learn from history.
			Interface anthropomorphism	Human likeness of interface in terms of appearance, language, and sounds etc.
Decision			Accuracy	Rate of accuracy of the decisions.
			Cost of decision	Money, efforts, and time involved with the decisions.
			Outlook	Gain or loss prospect of the decisions.
			Algorithm's role	Algorithm's role (supportive vs. competitive).
Delivery			Mode of delivery	The way decisions are provided (orally vs. electronic screen).
		Structural Characteristics	Explanation length	The length of explanations of the decision.
			Information quality	The quality of information provided in explanations.

themes of influencing factors of algorithm aversion, we then worked on the Excel data file related to this. To generate a first-order theme (T1), we reviewed all data and assigned one or more conceptual labels. Third, following an ontological process (Noy and McGuinness, 2001), first-order thematic labels were grouped based on similarity and interrelationship to form second and subsequent orders of themes (T2, T3, T4) when necessary. Fourth, core thematic areas (super theme) (T5) were identified, connecting with the topic of the current study. Fifth, derived themes (T1-T4) were summarized in Tables 4–6 and grouped by thematic area (super theme, T5), thus forming a taxonomic (sub-theme-super theme) hierarchy (Jones et al., 2011; Noy and McGuinness, 2001). A description of each theme was also presented (Jones et al., 2011). Sixth, a thematic map was constructed (Fig. 7) and compared with Tables 4–6 and spreadsheet data for consistency (Jones et al., 2011).

To ensure the quality of thematic analysis, we maintained some standard practices throughout this process. i) All studies were reviewed independently by the first and second authors; ii) After reviewing every 20 studies, findings were compared against each other; ii) Disagreement (if any) was resolved through discussion; iv) Each study was given equal attention; v) Themes were identified at a latent (interpretative) level going beyond a semantic (explicit) level (Braun and Clarke, 2006); vi) Themes were checked repeatedly against original data; vii) Consistency was ensured by iteratively matching data with ontology Tables 4–6 and the thematic map; viii) Themes were reexamined for internal coherence and uniqueness; ix) The process was thorough, rigorous, inclusive, and comprehensive; x) Insights from all authors were fully acknowledged.

4. Results

4.1. Study profiles

Reviewed studies were profiled to understand the status of the extant research on algorithm aversion.

4.1.1. Chronological view

Fig. 3 illustrates the year-wise distribution of the number of selected studies published from 1986 to March 2021. It appears that algorithm aversion research is still in its early stage and scholars have become interested in this topic very recently. Almost half of the studies ($n = 39$) emerged between 2018 and 2021, whereas the rest of the studies ($n = 41$) were published between 1986 and 2017. This surge in recent years could be attributed to the seminal work done by Dietvorst et al. (2015), who coined the term "Algorithm Aversion" (Berger et al., 2021). In our corpus of primary studies, 44 reviewed studies were published after the work of Dietvorst et al. (2015), out of which 75% ($n = 33$) studies cite this work, making it the most influential study in this area.

4.1.2. Publication venues

Identifying the publication venues of a given topic can potentially be beneficial for researchers who want to explore the topic more (Zahedi et al., 2016). The descriptive analysis reflects that 80 included studies were published by 19 publishers, led by Elsevier ($n = 24$), SAGE ($n = 11$), Wiley-Blackwell ($n = 9$), and Taylor & Francis ($n = 8$) (Appendix C). There were 9 publishers that each published only one study. The journals were distributed across different disciplines such as information technology/human-computer interaction ($n = 23$), business and economics ($n = 10$), psychology ($n = 7$), and decision science ($n = 6$) (Fig. 4). Journals that published studies related to the information technology/human-computer interaction discipline are the most favored venues, accounting for 40% of the studies ($n = 32$) (Fig. 4). We found that researchers with varied expertise (Fig 5) were interested in exploring this topic, suggesting the influence of this topic across different disciplines. This topic also testifies to a significant interdisciplinary collaboration; 17 studies (21.25%) are written in collaboration with authors from different disciplines. It is important to note that more than half of the authors (56.58% of all authors) are from the USA.

4.1.3. Research methodologies

Overall, experimental research design was the most preferred research method used in our reviewed studies. Sixty-nine studies (86.25%) were experimental study, 7 studies were survey studies (56,307 subjects), and 5 studies were field studies (64,184 subjects) (Fig. 6). Two studies were both experimental and field studies and one study was both experimental and survey-based. In 45 studies, researchers conducted more than one experiment. Altogether, 181 experiments were conducted on a total of 39,230 subjects. In designing experiments, researchers used incentivized experiments in 77% of instances ($n = 53$). Between-subjects design was used in 21 studies, while within-subject design was used in 4 studies. Twenty-one studies deployed both between-subject and within-subject designs in their studies. These experiments used a variety of data collection tools such as

Table 5
Supertheme: Individual; sub-themes and theme description.

T4	T3	T2	T1	Theme description
Pshychological Factor	General aversion	General distrust	Negative perception	General negative perception about algorithms.
			Lack of emotional trust	Lack of general trust in algorithms.
			Egocentric trimming	General tendency to prioritize one’s own decisions over others.
			Fear of change	Fear of changing behavior due to algorithm adoption.
			Perceived expectation	Preconceived expectation of perfect or near perfect decisions from algorithm.
			Perceived responsibility	People’s feeling of responsibility for their tasks?
			Moral obligation to follow own decision	People's feeling of moral obligation to follow their own decisions.
			Motivation	Incentive to follow algorithm.
Personality	Core self-evaluation factors	Self-efficacy	Self-esteem	Individual’s feeling of self-acceptance and self-respect.
			Expertise	User’s expertise in decision making.
			Online self-efficacy	Individuals’ perception about their ability to protect their data while online.
			Locus of control	Individuals’ degree of control over events of their lives.
		Neuroticism	Ability to use	Ability to use algorithms.
			Relationship concerns	Individuals’ concerns about relationship with human expert.
			Privacy concerns	General concerns about how personal data are collected and used.
			Extraversion	Describes how much a person is extrovert or introvert.
			Judger vs. perceiver	People’s way of making decisions.
Familiarity	With algorithm	Serial position effect (primacy vs. recency)	Status quo bias	Status quo resulted from previous use of algorithms.
			Affect heuristics	Heuristics developed from prior negative experience.
			Algorithmic error	Stage of decision-making when algorithmic error is observed.
			Introduction of algorithm	Stage of decision-making when algorithms are introduced.
			Length of familiarity	Length of familiarity with algorithms.
			Frequency of use	Frequency of use of algorithm.
			Awareness about algorithm's expertise	The level of awareness people has about algorithm's expertise.
			Training on algorithm	Training on algorithms that users have.
			With task	People's familiarity with task in hand.
			With human decision-maker	People's familiarity with human decision-maker.
Demography			Age	Age of people.
			Gender	Gender of people.
			Education	Level of education of people.

Table 6
Super themes: Task and High-level; sub-themes and theme description.

T4	T3	T2	T1	Theme description
Task		Nature of Task	Subjectivity	Tasks that require subjective judgement.
			Moral task	Tasks that is related to moral decision.
			Complexity	Task complexity (complex or simple).
High-level		Organizational	Profit orientation	Profit orientation of the organization.
			Oganization's type	Whether the company sells its own product or provides only decisions.
		Societal	Other people's views	Other people's views on algorithms and their users.
			Other users’ feedback	Use experience from existing or previous users.
		Environmental	Risk & uncertainty	Risks and uncertainty involved with the decisions.
		Cultural	Individualistic vs. collectivistic	Individualistic or collectivistic culture of macro environment.

observation of task performance, scenario/case/vignette, questionnaire, functional magnetic resonance imaging or functional MRI (fMRI), follow-up survey, open feedback, and interviews, either singly or in a combination of these. Though survey and field study were used in a relatively small number of studies, some of them were conducted on a large number of the population. For example, Castelo et al. (2019) conducted two online field studies on 56,000 participants to understand why and when consumers are likely to use algorithms. Similarly, Thurman et al. (2019) recruited 53,314 subjects to survey the attitudes towards news selection by algorithms. University students and staff (52.50%) were the most favored source of participant selection, followed by crowdsourcing platforms (38.75%) such as Amazon

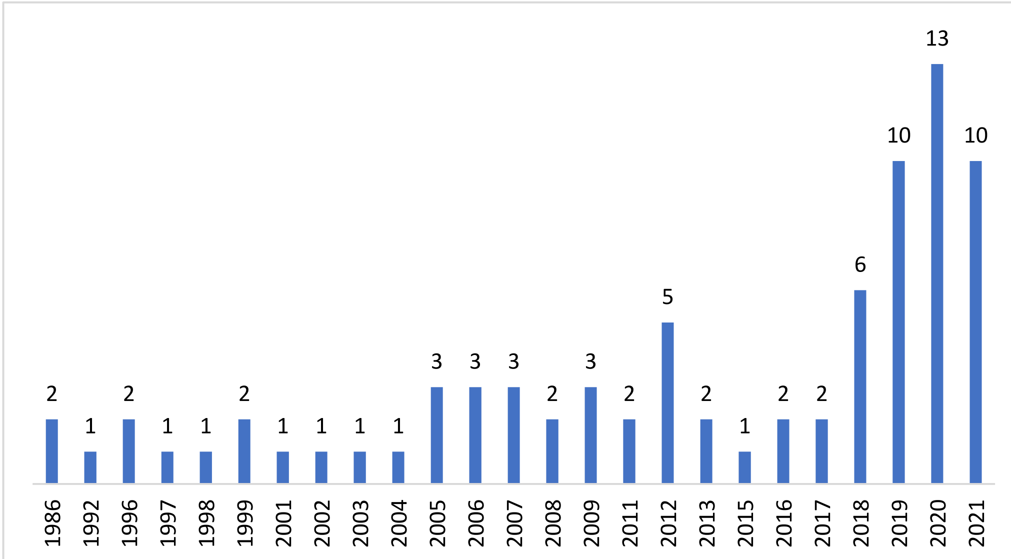

Fig. 3. Number of selected studies by publication year.

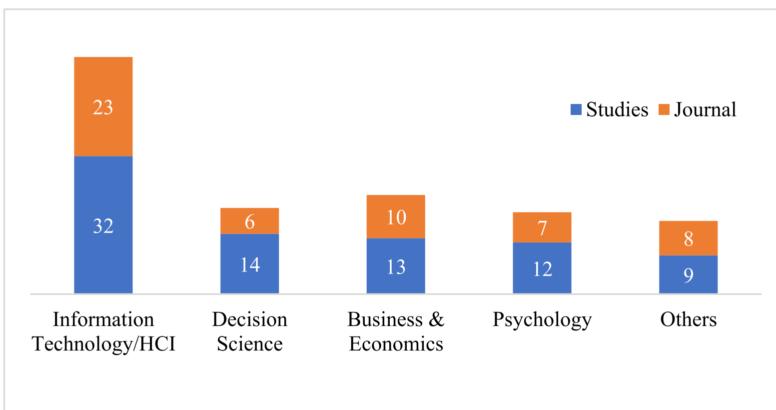

Fig. 4. Discipline of publication venue and studies.

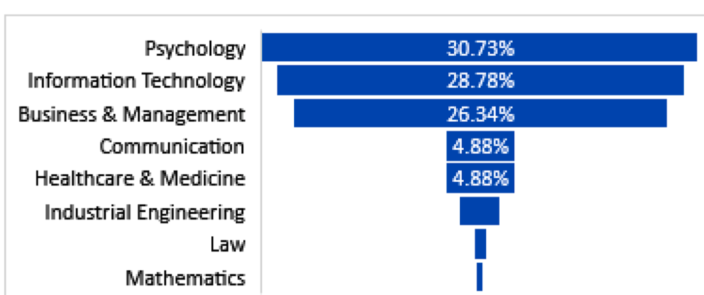

Fig. 5. Author's primary expertise.

Mechanical Turk (MTurk), Prolific, Online Recruitment System for Economic Experiments (ORSEE), SSI (Netherland), etc. A few studies used networking sites such as Facebook, LinkedIn, and WeChat, advertisements, local communities, and volunteers to select participants. Appendix D summarizes the details of each research method and design used, the characteristics of samples, foci of the studies, and the themes explored in the studies.

All 80 studies applied a quantitative data analysis technique, although a few studies (Litterscheidt and Streich, 2020; Park et al., 2019) used both qualitative and quantitative research methods.

4.1.4. *Research foci*

Over 50% of studies ($n = 41$) discussed algorithm aversion, while 13 studies discussed algorithm appreciation. A good number of studies ($n = 26$) investigated both algorithm aversion and appreciation. More than half of the studies ($n = 43$) were conducted in the context of the USA. Finance and healthcare sectors are explored more frequently, followed by judiciary, sales and marketing, personnel selection, and sports and

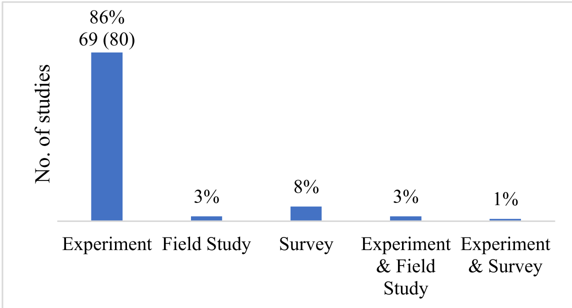

Fig. 6. Research method used in primary studies.

entertainment. The most investigated decision type is business decisions, followed by medical and judiciary decisions. A few studies also focused on moral (Bigman and Gray, 2018) and emotional (Prahl and Swol, 2021) decisions generated by algorithms.

4.2. *Influencing factors of algorithm aversion*

The primary goal of this SLR is to identify the influencing factors of algorithm aversion through critically evaluating the reviewed studies. The thematic map (Fig. 7) illustrates different themes concerning influencing factors of algorithm aversion, which emerged from our review. It corresponds to Tables 4–6, each of which comprehensively presents sub-themes and their descriptors. We present 56 first-order themes that emerged from the studies along with our interpretation under four super themes: algorithm, individual, task, and high-level.

4.2.1. *Algorithm factors*

Algorithm factors are involved with the factors that are related to the characteristics of decision design, presentation of decision, and the decision itself. Algorithm factors are shown in Table 4.

4.2.1.1. Design factors. Design factors include factors that are related to the design of algorithms, resulting in aversion. Several studies investigate such design factors and their impacts on algorithm aversion. One salient design factor is the "*black box" nature* of design, that is, lack of transparency. People exhibit aversion when they do not have a clear understanding of how the algorithm works and how it generates decisions (Dzindolet et al., 2002; Kayande et al., 2009). People are intrinsically interested in knowing the underlying principles of algorithmic decision making. Several studies indicate that algorithms may benefit from opening the "black box" (Litterscheidt and Streich, 2020; Sharan and Romano, 2020). The black box can be unpacked by making algorithms transparent. Transparency, in turn, can be achieved by making algorithms accessible, explainable, understandable, interactive, and integrative (Chander et al., 2018).

People prefer human decision makers because they have *access* to them and can ask about their reasoning behind the decisions (Önkal et al., 2009). Conversely, people rely less on algorithms because they do not have access to the algorithms' reasoning (Kayande et al., 2009; Önkal et al., 2009; van Dongen and van Maanen, 2013). Absence of access to algorithms' rationale makes people trust them less (Goodwin et al., 2013; Önkal et al., 2009). People seek justification when they receive decisions from algorithms (Lu et al., 2017). This is especially sought when there arises a cognitive dissonance due to nonconforming decisions (Festinger, 1957). However, people lack this scope due to the "black box" nature of algorithms' design, causing them to avert algorithms (Önkal et al., 2009). This problem could be ameliorated by providing *explanations* of the decision. Explanation can help users to understand the underlying rationale of the decisions (Gönül et al., 2006). When decisions come with explanation of how the algorithm works, people perceive it as more trustworthy (Goodwin et al., 2013; Zhang et al., 2021). However, it is not enough for an algorithm just to explain its underlying mechanism; explanation must be *understood*. By comparing computer recommendations to human recommendations of funny jokes, Yeomans et al. (2019) found that algorithm aversion stems not only from the content of the decisions but also from whether people can understand the algorithms. Henceforth, the decisions and explanations need to be communicated in persuasive ways such as using

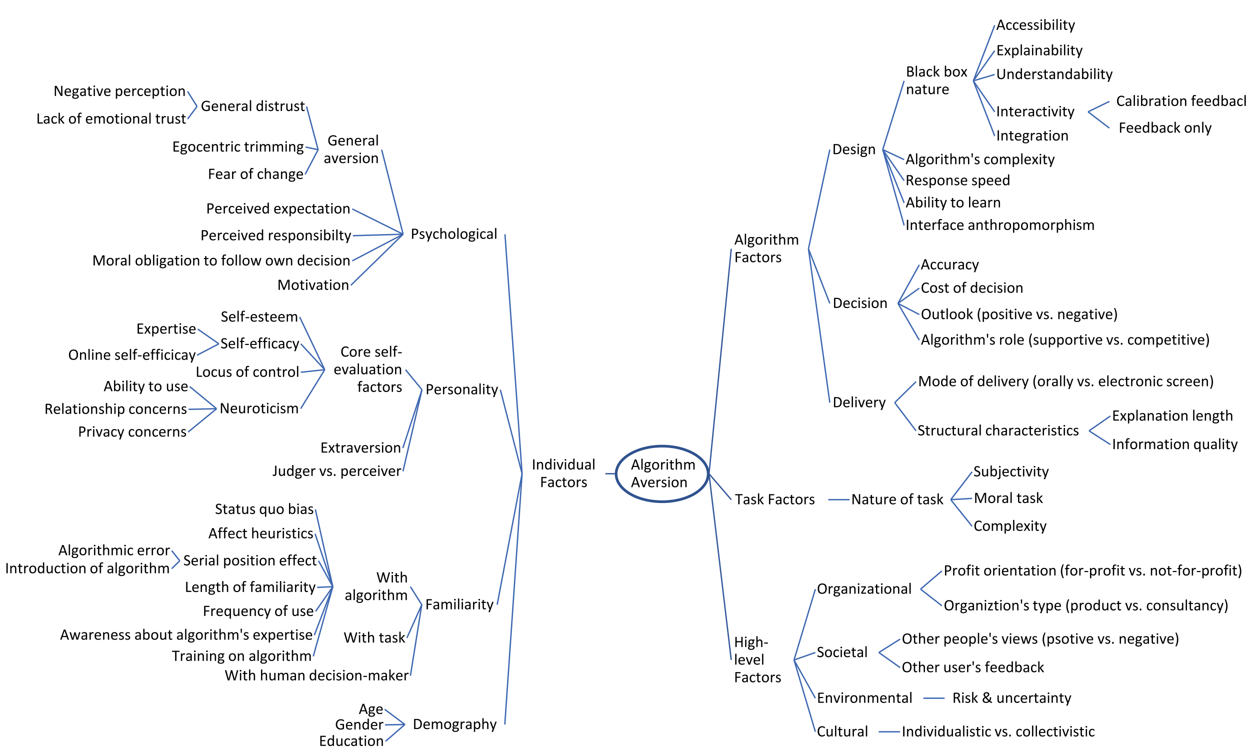

Fig. 7. Thematic map of algorithm aversion.

personalized conversation, using warm and kind words (Yun et al., 2021), providing illustrations (Zhang et al., 2021), and using a persuasive style of language (Önkal et al., 2019).

Moreover, people may not have a dataset to validate the algorithmic decisions against. Therefore, it is important that people can interact fluidly with the algorithms, more specifically modifying the input data in response to algorithms' feedback so that they can validate the decisions through trial-and-error processes (van Dongen and van Maanen, 2013). This *interactive* process may continue iteratively until individuals reach their satisfaction level. Such *calibration process* reduces overconfidence (Sieck and Arkes, 2005) and improves the calibration of trust (van Dongen and van Maanen, 2013). Supporting this finding, Sharan and Romano (2020) point out that if people are given the option to modify the algorithm—even slightly—it may increase trust in algorithms (Dietvorst et al., 2018). This elicits a sense of satisfaction and fulfills the desire for control (Dietvorst et al., 2018). However, if *only feedback* about the performance of humans' decisions, algorithms' decisions, or both is provided without any iterative process, people's reliance on algorithms may decrease. This is because having feedback elicits "its own set of heuristics and biases," stemming from people's tendency to overestimate their performance compared to others (van Dongen and van Maanen, 2013). Furthermore, since negative experiences have a greater influence than positive experiences (van Dongen and van Maanen, 2013), people tend to follow algorithms less if they receive feedback about algorithms' poor performance (Dietvorst et al., 2015).

Involving individuals and *integrating* their opinions in algorithmic decision processes reduces aversion (Köbis and Mossink, 2021). Integrating opinions helps people to monitor and adjust an algorithm and its outcome (Köbis and Mossink, 2021). It also helps algorithm designers to understand the personality traits of individuals, which eventually assist them to develop algorithms, matching individuals' profiles (Nass and Lee, 2001). Such integration also plays a role in ameliorating the psychological reactance of individuals, stemming from decision incongruity between one's own decisions and the algorithm's (Fitzsimons and Lehmann, 2004). People tend to believe that each decision task is for a special purpose (Goodwin et al., 2013) and that algorithms suffer from reductionism, failing to consider unique situations (Newman et al., 2020). More specifically, people think that algorithms consider many factors in general but neglect certain qualitative factors and unique contexts of the decisions (Langer and Landers, 2021), and that humans are able to consider these. Such a perception leads to decision incongruity. Integrating people's opinions can reduce such decision incongruity, as they bring the vision, purpose, and context of the task into algorithmic decision-making. Additionally, people feel more confident and become more committed to complying with algorithms when their opinions are integrated (Kawaguchi, 2021; Landsbergen et al., 1997).

Apart from the "black box" nature of their design, several other design issues may influence algorithm aversion. For instance, research has found that people are more aversive of decisions that are based on *complex algorithms* than on simple algorithms (Stein et al., 2020). They perceive complex algorithms as "eerie and highly unnatural" and a threat to "human uniqueness" (Stein et al., 2020). Again, people attribute much of their aversion to the *speed of algorithms' response*. People consider slowly generated decisions by algorithms less accurate and thereby rely on them less, although they think the opposite in the case of decisions provided by humans (Efendić et al., 2020). They see decision tasks as simple and less travailing for algorithms. Therefore, they expect algorithms to be faster in generating decisions. However, this finding is inconclusive. In one study, Park et al. (2019) found that people are likely to follow an algorithm if it responds slowly. People use this waiting time to mull over the task and the decision process of themselves and the algorithm. Such reflection helps them to reexamine the two processes and compare them carefully. Nevertheless, it seems intuitive that a fast response has much more positive impact than a slow response, as speed of response is related to the efficiency of decision-making. We support this intuition by our observation of increased tendency among researchers to make algorithms' responses faster (Cheng et al., 2012; Daming et al., 2020; Zhang et al., 2009; Zhu and Wang, 2004)

People often reject algorithms, seeing them err and believing that algorithms are unable to learn from their mistakes (Dietvorst et al., 2015). In this regard, demonstrating algorithms' *ability to learn* may alleviate algorithm aversion (Berger et al., 2021). Such ability can be demonstrated by incorporating ML in algorithms' design (Lennartz et al., 2021; Li et al., 2020). Finally, our review finds that algorithm aversion is greatly influenced by its *anthropomorphic design of interface* (Castelo et al., 2019; Madhavan and Wiegmann, 2007; Pak et al., 2012; Zhang et al., 2021). Interface anthropomorphism can be of several forms: how humanlike the interface looks, the language it uses, the sounds it makes, and the presence of liveliness (Pak et al., 2012; Stein et al., 2020). It can be designed in several ways such as using human voice-based communication (Qiu and Benbasat, 2008), instant chat/avatars (Zhang et al., 2021), and a photo of an expert (Pak et al., 2012). Anthropomorphic design helps to establish a social presence, thus increasing the acceptance of algorithms (Qiu and Benbasat, 2008; Zhang et al., 2021). However, anthropomorphistic design needs to be user-friendly to have the intended effect (Li et al., 2020).

4.2.1.2. Decision factors. Our review finds several influencing factors related to the characteristics of the decision: accuracy, cost, outlook, and roles it plays in decision-making. First, *decision accuracy* is a key factor of algorithm adhesion. People punish bad decisions generated by algorithms more severely than those generated by humans (Bogert et al., 2021). People lose confidence in algorithms when they produce inaccurate decisions (Bogert et al., 2021; Dietvorst et al., 2015). Such inaccuracy has serious consequences if it occurs in an easy task. Individuals' level of trust decreases significantly when they observe algorithms failing to provide accurate decisions for an easy task, believing that algorithms would not be capable of solving more difficult tasks (Madhavan et al., 2006). Second, people consider the *cost of the decisions* in deciding whether they follow algorithms. People follow paid decisions significantly more than freely available decisions (Gino, 2008). For example, in online recommendations, which come without incurring any direct cost, people do not exhibit much interest in abiding by these decisions (Sutherland et al., 2016). On the contrary, if people need to spend money, time, and effort to receive decisions as it happens in the case of using a corporate decision support system, for the sake of honoring sunk costs, they are unwilling to forego the decisions (Sutherland et al., 2016).

Third, *outlook of the decisions*, that is, whether the decisions lead individuals to gain or loss, dictates algorithm aversion. For instance, Toma et al. (2020) performed a between-subject experiment to examine the impact of prospects (gain and loss) on the reliance of automated decisions and found that individuals rely more on algorithms when the forecasted decisions led them to gain. Similarly, Kayande et al. (2009) found that upside potential motivates people to exert greater efforts to follow an algorithm. However, it may be riskier to generalize this finding as people who are relatively conservative and risk-averse may favor algorithms that predict a negative outlook. For example, Swinney (1999) found that auditors, who are conservative by profession, show an overreliance on algorithms that project a negative outlook.

Finally, in some instances, the *role of algorithmic* decisions in relation to human decisions is a crucial determinant of algorithm aversion. For example, people tend to rely less on algorithms when they are meant for replacing human decisions rather than supporting human decisions. People prefer human decisions empowered by algorithms and rely more on algorithmic decisions so long as algorithms are subordinate to humans, and humans make the final decisions (Bigman and Gray, 2018; Zhang et al., 2021)

4.2.1.3. Delivery factors. Factors related to decision delivery consist of *mode of delivery* and presentation styles of decision. Önkal et al. (2009) suggested that people prefer human decisions to algorithmic decisions because human decisions are typically delivered orally, which demands attention. On the other hand, algorithmic decisions are presented through an electronic screen, which does not draw as much attention. Therefore, algorithmic decisions may be provided orally through human-like agents. However, delivering decisions orally is not alone sufficient to overcome aversion. The agents need to be realistic. For instance, Laakasuo et al. (2021) found that decisions delivered by a mere human-looking robot instead of robot with a clear robotic appearance result in aversion. Again, *structural characteristics* of presentation of decisions affect individuals' willingness to follow automated decisions (Gönül et al., 2006). By structural characteristics, we refer to the *length of the explanation* and the *quality of information* of the explanation. In an experiment of time series plots with short (1–1.5 lines) and long (4–5 lines) explanation with varied degrees of information value, Gönul et al. (2006) found that long explanations and explanations with higher information value are more persuasive.

4.2.2. Individual factors

Individual factors related to algorithm aversion consist of a wide range of factors including psychological factors, personality traits, demography, and individuals' familiarity with algorithms and tasks. By psychological factors, we imply those factors related to individuals' reasoning, logic, thinking, and emotion in connection to algorithmic decisions. Personality trait factors include factors from self-evaluation and Big Five factors. Familiarity factors denote individuals' experience with algorithms, tasks, and human advisors. Finally, demographical factors refer to the influence of gender, age, and education on algorithm aversion. The individual factors are shown in Table 5.

4.2.2.1. Psychological factors. Our review finds that some people are habitually averse to algorithmic decisions without judging the merit of such decisions, which we call *general aversion* (Kawaguchi, 2021; Önkal et al., 2009; Prahl and van Swol, 2017). Such general aversion could be attributed to the *general distrust* that stems from *negative perceptions* people hold about algorithms (Shaffer et al., 2012) or from *lack of emotional trust* (Zhang et al., 2021). Workman (2005) found that people's perceptions and attitudes towards algorithms are strongly associated with algorithm aversion. In a field experiment on more than 6000 customers about the effectiveness of sales calls generated by either algorithms or humans, it was revealed that when customers are aware of the conversational partner being an algorithm, they purchase less because they perceive algorithms as "less knowledgeable and less empathetic" (Luo et al., 2019). Consistent with this, Berger et al. (2021) also found that people prefer human decisions to algorithmic decisions if they know the source of the decisions. Similarly, by examining stock price forecasting, Önkal et al. (2009) showed that this general distrust prevails even when there is no clear indication about the superiority of human decisions over algorithmic decisions. Interestingly, people extrapolate this negative perception to the user of the algorithms as well. People think that individuals who use algorithmic aid are less intelligent (Prahl and van Swol, 2017). Furthermore, the *degree of emotional trust* they invest in algorithms has a great impact on algorithm aversion. Lack of emotional trust, such as feeling secured, comfortable, and contended, dissuades people from welcoming algorithmic decisions (Zhang et al., 2021). Negative emotion results in lower utilization of algorithms (Gogoll and Uhl, 2018; Prahl and Swol, 2021; Promberger and Baron, 2006). People derive negative emotion from the belief that machines are not meant to perform decision tasks (Prahl and Swol, 2021). Echoing the finding that people suffer from a lack of emotional trust in algorithms, Lennartz et al. (2021) empirically showed that in case of treatment decisions, people prefer to double-check the algorithmic decisions with a physician. They also demonstrated that in case of disagreement between physician and algorithms, most people prefer to rely on a physician's decision, a phenomenon that can be attributed to the lack of emotional trust in algorithms. Another reason for general aversion can be the human nature of *egocentric trimming*, which implies a general tendency to prioritize one's own decisions over others (Sutherland et al., 2016; Yaniv and Kleinberger, 2000). This indicates that people approach algorithmic decisions with a predilection to not completely adhere to them (Sutherland et al., 2016). Furthermore, general aversion can arise due to *fearing the risk of changing behavior* because of the adoption of algorithmic decisions (Kawaguchi, 2021). However, a few studies found an absence of general aversion (Köbis and Mossink, 2021). For example, Prahl et al. (2017) found no general aversion in the context of operating room management decision tasks in hospitals.

Next, people's *perceived expectation* from algorithms determines the use of algorithmic decision. People expect algorithms to make fewer mistakes and be more accurate than human experts (Madhavan and Wiegmann, 2007). They have a preconceived notion of algorithms becoming perfect or near-perfect ("perfect automation schema") (Dzindolet et al., 2001), and perceive algorithms as experts (Goodyear et al., 2017; Li et al., 2020). Therefore, they set higher initial expectations about algorithms, resulting in bias towards algorithms. This bias increases the tendency of individuals to notice and remember errors made by algorithms (Dzindolet et al., 2002). Therefore, any algorithmic error observed by individuals makes them underestimate the perceived utility of algorithmic decisions and eventually abandon it (Dzindolet et al., 2002). However, this preconceived notion about "perfect automation schema" is debatable. Dietvorst and Bharti (2020) found that people believe that it is humans rather than algorithms having the highest likelihood of providing perfect or near-perfect decisions. Algorithmic decisions are more consistent, maintaining constant average performance, and thus have lower variance and lower possibility of generating perfect score. On the other hand, human decisions exhibit higher variability with a higher likelihood of near-perfect decisions, and with the corresponding downside of being more likely to generate inaccurate decisions (Dietvorst and Bharti, 2020).

Again, feeling responsibility for the consequences of decisions and a moral obligation to follow their own decisions may create algorithm aversion. The less people *perceive responsibility* for the task, the less they are willing to exert mental effort to accept algorithmic decisions (van Dongen and van Maanen, 2013); rather they feel comfortable with human decisions (Promberger and Baron, 2006). It was found in a visual detection experiment by Dzindolet et al. (2002) that another psychological factor that is responsible for algorithm aversion is individuals' feeling of *moral obligation* to follow their own decisions and not the decisions of others.

Research has demonstrated that people follow algorithms if they are *motivated* to follow them. People feel motivated when pecuniary and non-pecuniary incentives are associated with following algorithms (Lim and O'Connor, 1996; van Esch et al., 2020). Incentives increase confidence among users to accept algorithms (Köbis and Mossink, 2021). This finding is also supported by the fact that in 77% of our selected studies, researchers employed incentivized experiments given their salient impact on algorithm aversion. Incentive is also crucial for continued and effective use of algorithms. If there is a lack of incentive, people do not feel interest to exert mental effort to evaluate the efficacy of algorithmic decisions; thereby, they often favor human decisions (Dijkstra, 1999). However, incentives do not always have impact on aversion. In an experiment, Köbis and Mossink (2021) found that following algorithms is not only a matter of incentives but also the ability of the individuals. Interestingly, another study by Arkes et al. (1986) showed that when people are incentivized not to rely on algorithms, they do not follow algorithms.

4.2.2.2. Personality factors. In connection to algorithm aversion, our review finds several factors related to the "core self-evaluation" concept and Big Five personality traits. *Core self-evaluation* refers to "fundamental, subconscious conclusions individuals reach about themselves, other people, and the world" (Judge et al., 1998). According to Judge et al. (1997) individuals' appraisals of the external world are influenced not just by the characteristics of objects and individuals' attitudes in relation to those objects but also by the attitudes individuals hold about themselves, other people, and the world. Judge et al. (1998) proposed four self-evaluation factors: self-esteem, self-efficacy, locus of control, and neuroticism, which we find relevant to algorithm aversion.

Self-esteem refers to an individual's "feeling of self-acceptance and self-respect" stemming from "subjective evaluation of his or her worth as a person" (Orth and Robins, 2014). Our review finds multiple studies that reported algorithm aversion due to the feeling of self-esteem of individuals. In an experimental study regarding personnel hiring based on algorithmic decision, Lee (2018) showed that job candidates find it demeaning and dehumanizing to be judged by a machine. Additionally, when people come across a decision that contradicts their own decisions, they experience a sense of psychological reactance (for more see Brehm, 1966), which in turn discourages them from following algorithms (Fitzsimons and Lehmann, 2004). In addition, egocentric bias, which is favoring one's own decisions over other's, allures individuals to choose their own decisions rather than algorithmic decisions (Logg et al., 2019). They are willing to accept the outcomes of their own bad decisions, not that of algorithms (Sutherland et al., 2016).

Self-efficacy implies "people's beliefs in their ability to influence events that affect their lives," and the most efficient way of developing self-efficacy is through learning from experience (Bandura, 2010). Extant literature has demonstrated that experienced people, who feel a greater degree of self-efficacy and who are more experienced, rely less on algorithmic decisions (Arkes et al., 1986; Logg et al., 2019; Önkal et al., 2019; Whitecotton, 1996). They feel greater confidence in their ability and are more likely to rely on their own decisions (Sharan and Romano, 2020; Sieck and Arkes, 2005). They feel less confident in algorithmic decisions. Therefore, they exert less commitment to abide by algorithmic decisions, resulting in negligence of algorithms (Landsbergen et al., 1997). Performing an experiment on *expert and lay* people about business and geopolitical decision-making tasks, Logg et al. (2019) showed that expert people, who possess professional or specialized knowledge in a given subject, rely less on algorithmic decisions than lay people do. Interestingly, when this self-efficacy is considered from the lens of people's behavior online, that is, *online self-efficacy,* which denotes "individuals' own assumptions about their level of ability to protect their data" (Araujo et al., 2020), people's reaction to algorithmic decisions is reversed. With a higher degree of online self-efficacy, people feel more confident in using algorithms (Araujo et al., 2020). More specifically, the greater individuals' belief about their online self-efficacy, the higher the positive attitudes individuals have towards the utilization of algorithmic decision.

Locus of control concerns individuals' beliefs that they can influence the events of their lives (internal locus of control) or that environment or fate dictates the events of their lives (external locus of control) (Judge et al., 1998). People desire to retain their control in decision-making by making decisions by themselves (Dzindolet et al., 2002). Our review finds evidence that desire for control inhibits individuals from following algorithmic decisions. In medical diagnostic decisions, it is observed that greater internal locus of control is positively associated with algorithm aversion (Shaffer et al., 2012). Similarly, Kawaguchi (2021) found that when people are entrusted with a higher degree of task delegation, which in turn gives them greater control over the task, they follow algorithmic decisions less. However, this may not always necessarily mean that people want full control over decision process. People might become satisfied even with slight control over the process, such as the ability to modify the input to some extent in algorithmic decision-making (Dietvorst et al., 2018).

Neuroticism, also a Big-Five personality factor, is related to people's feelings of anxiety, insecurity, guilt, and timidness (Judge et al., 1998). Neurotic people are fearful of novel situations (Judge et al., 1998), resulting in aversion towards algorithms. Complementing this notion, Sharan et al. (2020) argue that people with high neuroticism perceive algorithms as less trustworthy. While by using algorithms, just as any new technology, some people feel a sense of accomplishment and being innovative, others feel anxiety to comply with algorithms (van Esch et al., 2020). Research suggests that the more people feel anxiety about using technology, the higher their aversion to technology (Meuter et al., 2003). Anxiety may arise from people's concerns about their *capability of using algorithms*. In an experiment regarding personnel selection, van Esch et al. (2020) found a negative relationship between the anxiety about the capability of using algorithms and algorithm acceptance. People also become anxious about the potential impact of algorithmic decisions on the *relationship with the human expert* that people build. People value their personal relationship with the human expert and view that with the advent of algorithmic decision-making, this relationship might be jeopardized. A similar notion is speculated by Shaffer et al. (2012), who argue that patients derogate physicians using decision aids because they have concerns about the effect of decision aids in patient-clinician interaction. Finally, people's perceived *privacy concerns* affect their reliance on algorithmic decisions (Vimalkumar et al., 2021). Privacy concerns refer to the "general concerns about how personal data are collected and used" (Araujo et al., 2020). Several prior studies agree that there is a positive association between privacy concerns and algorithm aversion (Araujo et al., 2020; Thurman et al., 2019). More specifically, the higher the privacy concerns the individuals have, the lower the utilization of an algorithm.

The degree of *Extraversion* also influences people's aversion to algorithms. Extraversion, a key Big Five personality trait, describes to what degree a person is an extrovert or introvert. Extroverted people are characterized by sociability, assertiveness, and excitability, whereas introverted people are characterized by quietness and shyness (McCrae and Costa, 1987). Extroverted people are risk-seeking and motivated to gamble with their decisions (Dietvorst and Bharti, 2020). They tend to pursue decisions that have the possibility of highest positive outcome, disregarding the corresponding negative prospect (Dietvorst and Bharti, 2020). They possess higher sensitivity to decision errors made by algorithms, causing them to rely more on humans over algorithms, believing that humans have the highest likelihood of making near-perfect decisions (Dietvorst and Bharti, 2020; Merritt and Ilgen, 2008).

Finally, the degree of algorithm aversion can be different depending on *judging and perceiving* personalities of individuals (Myers et al., 1998). Perceivers tend to analyze cost-benefits while making decisions and seek inner meaning and contributions thereon. On the other hand, judgers tend to make decisions based on their gut feelings and intuition. In an experiment on nursing staffs about algorithm adoption, McBride et al. (2012) found that judgers are less willing to use algorithms than perceivers.

4.2.2.3. Familiarity. The current study finds that people's familiarity with algorithms, tasks, and human experts has significant impact on decision aid reliance. Extant literature has demonstrated that lack of *familiarity with algorithms* leads to higher aversion to algorithms (Lim and O'Connor, 1996). In other words, familiarity with algorithms develops a sense of *status quo* bias among individuals, which results in algorithm acceptance (Fenneman et al., 2021). Nevertheless, becoming familiar with algorithms could be a double-edged sword dilemma. While familiarity with algorithms reduces algorithm aversion (Whitecotton, 1996), it may be counterproductive if people experience bad decisions and error with algorithmic decisions. Negative experiences with algorithms lead to more serious consequences for algorithm utilization. Such a negative experience develops a sense of *affect heuristics* (Liu et al., 2019; Slovic et al., 2004), which negatively impacts on people's attitudes towards algorithms (Li et al., 2020). Several studies on investment (Fenneman et al., 2021; Whitecotton, 1996), demand management (Berger et al., 2021; Feng and Gao, 2020; Yuviler-Gavish and

Naseraldin, 2016), healthcare (Prahl and van Swol, 2017), student success (Dietvorst et al., 2015), and safety (Liu et al., 2019) and security systems (Merritt and Ilgen, 2008) corroborate this finding. Negative experience causes individuals to feel regret for complying with algorithms. The more individuals feel such post-decision regret, the higher their tendency to avoid algorithms (Feng and Gao, 2020; Hung et al., 2007; Kawaguchi, 2021). Moreover, the stage of the decision-making process at which algorithmic error is observed is important. The degree of aversion varies depending on whether the error occurs at the early stage or the later stage of decision-making. According to *serial--position effect* theory (Colman, 2015; Ebbinghaus, 1913), primacy effect is stronger than recency effect (Worchel et al., 1988). Therefore, *algorithmic error* that occurs at the early stage of decision-making (primacy effect) has more negative effect than the error that occurs at a later stage (recency effect) (Manzey et al., 2012). The same serial-position effect could also be applied to whether *algorithmic decisions are introduced* in the early stage or later stage of a decision-making process. Having algorithmic decisions at the early stage of one's own decision-making process increases the likelihood of algorithm utilization (Dijkstra, 1999). Consistently, Yuviler-Gavish and Gopher (2011) found that people's trust increases if they are familiarized with some simulated decisions before the final decisions. However, this finding is nisi and requires scrutiny as other researchers argue the opposite. For instance, van Dongen and van Maanen (2013) showed that people's trust in algorithms increases if algorithmic decisions are presented once people have made their own decisions.

Interestingly, in some cases, mere familiarity may not be sufficient for the continued use of an algorithm. *How long* people have been familiar with an algorithm and *how frequently* they use it are also important. Prahl and van Swol (2017) found that in hospital management, people trust algorithms less because they have a relatively short period of experience with them. Additionally, as the rate of use increases, people are likely to experience bad decisions more frequently from algorithms, instigating higher aversion to algorithms (Merritt and Ilgen, 2008).

Algorithm aversion is also influenced by people's *awareness about algorithms' expertise* and efficiency. Research has found that awareness of algorithms' expertise, accuracy, and other salient attributes motivates people to rely more on algorithms (Bigman and Gray, 2018; Sutherland et al., 2016; Zhang et al., 2021). The greater the perceived expertise of algorithms people have, the lower the likelihood of algorithm aversion (Gunaratne et al., 2018; Pearson et al., 2019). Further, a lack of knowledge about how to use algorithms hinders the usage of algorithms (MacKay and Elam, 1992). In this regard, *training* on how to use algorithms and how to recognize situations in which algorithms are likely to be useful can be helpful (Dzindolet et al., 2002; Green and Hughes, 1986). Research has found that individuals are motivated to use algorithms if they are trained in statistical techniques and algorithms (Araujo et al., 2020; Önkal et al., 2019). In a similar way, Litterscheidt and Streich (2020) argue that financial education increases the likelihood of acceptance of robo-advisor. In addition, hands-on experience with algorithms in simulated environments can also reduce the aversion to algorithms (Li et al., 2020; Luo et al., 2019).

Additionally, though *task experience* increases the self-efficacy of individuals, a few studies have found that familiarity with tasks does not have any impact on the reliance on decision aid (Kawaguchi, 2021; Whitecotton, 1996). Finally, people trust algorithms less if they are *familiar with human decision-makers*, whose decisions are compared with algorithmic decisions, and perceive him or her as an expert (Sharan and Romano, 2020; Thurman et al., 2019).

4.2.2.4. Demography. The degree of algorithm acceptance is varied depending on the age, gender, and levels of education of individuals. Our review finds that people from different *age* groups view algorithms differently. For example, older people perceive algorithmic decisions as less useful (Araujo et al., 2020) and therefore trust them less (Lourenço et al., 2020). In one experiment it was found that older people prefer news recommendations from human advisors more than those of algorithms (Thurman et al., 2019). However, this relationship is not unequivocal. For example, in another experiment with a medication management system, it was found that when it comes to deciding between individuals themselves and algorithms, older people showed a greater reliance on algorithms than younger people did because older people feel less confident in their ability (Ho et al., 2005). Intriguingly, in multiple experiments related to business and geopolitical decision-making tasks, Logg et al. (2019) found no such impact of age on algorithm aversion. *Gender* may also influence the perception of algorithms. Research in multiple domains such as media, health, and justice revealed that females see algorithms as less useful (Araujo et al., 2020). However, there is also evidence in the news and telecommunication industry that gender is not relevant to algorithm preference (Thurman et al., 2019; Workman, 2005). Furthermore, people appreciate algorithms less when their levels of *educational* achievement is lower and when they feel less comfort with numbers (Thurman et al., 2019; Logg et al., 2019).

4.2.3. Task factors

Whether the task requires subjective evaluation or moral judgment, or whether the task is simple or complex all influence people's attitudes on algorithm acceptance. The task-related factors are presented in Table 6.

People are reluctant to use algorithms for decisions that require feeling and thinking capability (Bigman and Gray, 2018). An algorithm is superior in performing mechanical and objective tasks but not subjective tasks. Algorithms lack intuition and *subjective* judgment capability (Lee, 2018). Therefore, people perceive them as less trustworthy and fair when they see algorithms performing tasks that require subjective evaluation (Castelo et al., 2019; Dijkstra et al., 1998; Lee, 2018). People obey algorithms less when subjective judgments are more important (Bogert et al., 2021; Longoni and Cian, 2020). Extant literature has found that there is a positive correlation between perceived objectivity and algorithm utilization (Castelo et al., 2019; Zhang et al., 2021). Task objectivity increases comfort with using algorithms and positive attitudes towards perceived effectiveness (Castelo et al., 2019). How task objectivity can be increased is contingent on how the task is framed and what components of tasks are prioritized. For example, personnel hiring can be done either through objective evaluation by using psychometric tests or through subjective evaluation by conducting interviews and relying on intuition (Castelo et al., 2019).

Again, people perceive *moral decisions* to be the domain of humans, not algorithms (Niszczota and Kaszás, 2020), and thus delegating moral decisions to algorithms as less favorable (Lee, 2018). As such, people intuitively turn away from algorithms when they see algorithms making moral decisions as well as decisions related to legal, medical, and military concerns (Bigman and Gray, 2018; Gogoll and Uhl, 2018). Finally, people discount algorithmic decisions when the task is relatively *simple* and does not require high computational skills (Önkal et al., 2009).

4.2.4. High-level factors

Beyond algorithm, individual, and task factors, the role of *organization*, which utilizes algorithmic decision, *societal factors, cultural context, and risk and uncertainty* involved with tasks, is also examined in the reviewed literature (Table 6). We call these organizational, societal, and cultural factors high-level factors (Parker and Grote, 2020). Our review finds that people have less trust in decisions provided by *for-profit firms* such as bank, insurance, and other private firms than in decisions provided *not-for-profit firms* such as government-run firms (Lourenço et al., 2020). Again, people are less satisfied with algorithmic decisions provided by firms that *sell their own products* than those that provide *only decisions* (Lourenço et al., 2020). However, our review finds no significant impact of firm size, industry, or product type on algorithm aversion

(Sanders and Manrodt, 2003).

Social influence, that is, *other people's views* (third-party, see Langer and Landers, 2021) on an algorithm and its users shapes how individuals perceive algorithms. People are greatly influenced by the perception of others such as colleagues, friends, and supervisors in algorithm adoption (Workman, 2005). People think of algorithms as less professional and fair, and therefore people who use it are perceived as less capable and intelligent (Arkes et al., 2007; Diab et al., 2011; Eastwood et al., 2012). Such views of society influence people unfavorably in algorithm utilization. Again, we find evidence that having *feedback from existing or previous users* of algorithmic decisions and information about the impact of these decisions on their performance determines people's willingness to follow algorithms (Alexander et al., 2018; Zhang et al., 2021).

Several studies have highlighted that if the decision domain and environment are *risky and volatile*, people reject even the best possible algorithms (Dietvorst and Bharti, 2020; Feng and Gao, 2020;

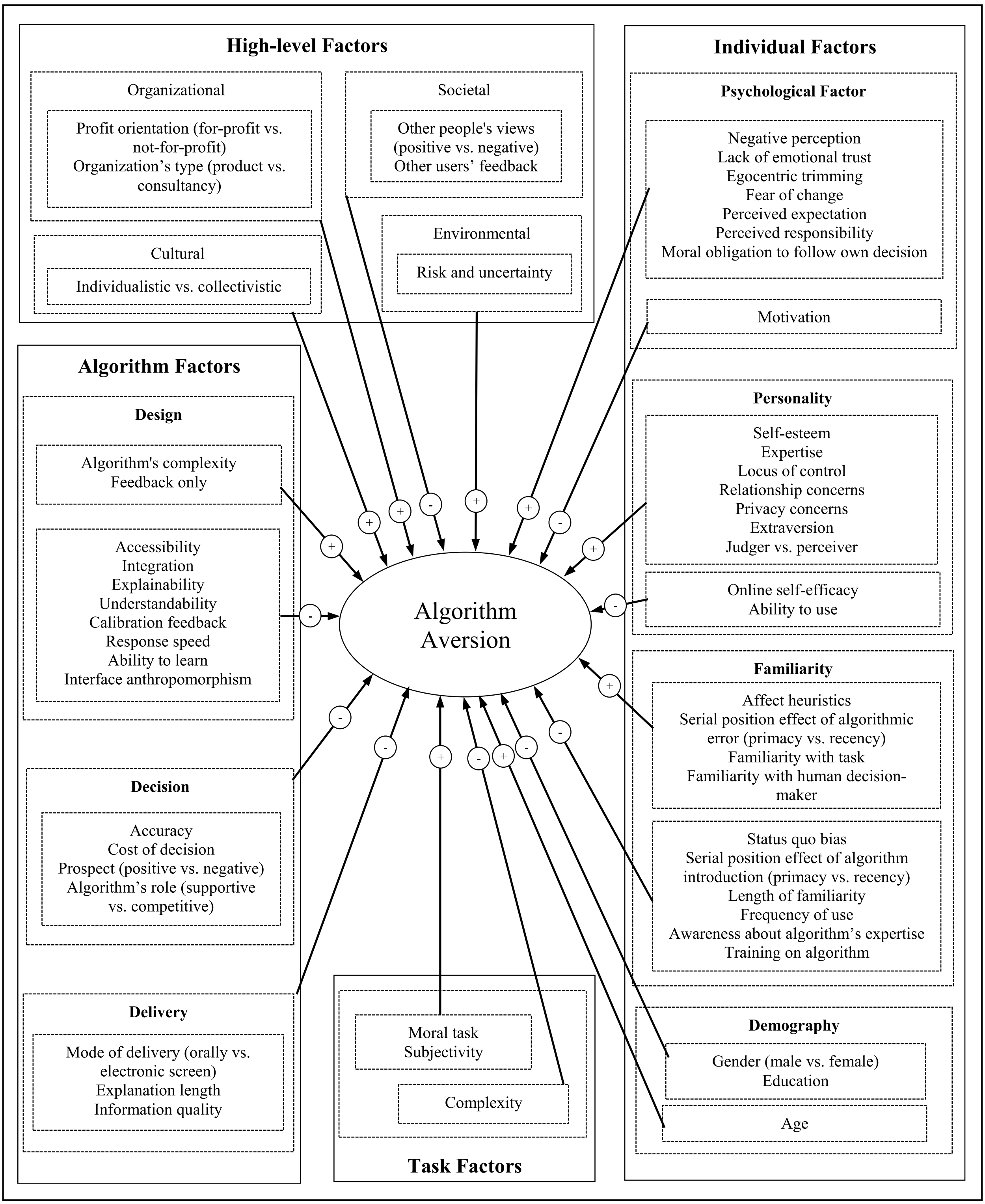

Fig. 8. Comprehensive framework for algorithm aversion and its factors explored in existing literature.

Kawaguchi, 2021; Sutherland et al., 2016; Zhang et al., 2021). For instance, comparing the perception of humans and robo-advisors in the high-risk financial domain, Zhang et al. (2021) showed that people prefer human advisors to algorithms. An identical result was also found in high-profit conditions in a newsvendor problem by Feng and Gao (2020). Again, in highly uncertain environments such as medical decision-making, demand forecasting, and predicting voter behavior, people tend to avoid algorithmic decisions (Dietvorst and Bharti, 2020; Kawaguchi, 2021). This may be due to people feeling insecure regarding the outcome and concern about the consequences of this outcome (Grgić-Hlača et al., 2019; Lennartz et al., 2021). Therefore, they are less willing to delegate such serious decisions to algorithm. However, this finding is not absolute. Based on an experiment on a war-like situation, Sutherland et al. (2016) argue that people use algorithms more in uncertain environments.

Lastly, *country-level cultural context* also influences an individual's attitudes towards algorithms. Culture is an important influential factor in algorithm adoption (Duan et al., 2019). For instance, Germans are relatively more *individualistic*, and they prefer to rely on individual decisions. Hence, they are less likely to follow algorithms. On the other hand, the Chinese are relatively more *collectivistic*, preferring collective decision. Therefore, they tend to prefer algorithmic decisions more (Rau et al., 2009).

5. A comprehensive framework

Insights from the present SLR study aid us in the development of a comprehensive framework for algorithm aversion, comprising all influencing factors explored so far (Fig. 8). The framework is comprised of factors under four major categories: algorithm, individual, task, and high-level. As discussed earlier, algorithm factors consist of design, decision, and delivery factors. Our review suggests that decision and delivery factors are negatively associated with algorithm aversion, meaning that people tend to prefer algorithmic decisions when the magnitudes of decisions and design factors are high. Again, while most of the design variables are negatively related to the development of algorithm aversion, factors such as algorithms' complexity and feedback-only mechanism are positively associated with algorithm aversion. We segregate individual factors into psychological, personality traits, familiarity, and demographical variables. Both personality traits and familiarity factors are positively related to algorithm aversion. However, in the case of psychological factors and demographical variables, the relationships with algorithm aversion are not definitive. While factors such as preconceived expectations, responsibility, anxiety, and age are positively related, other factors such as perception, motivation, and education are negatively associated with algorithm aversion. Task factors such as subjectivity and moral tasks positively influence algorithm aversion, whereas objectivity and complexity of tasks increase algorithm utilization. We group organizational, societal, environmental, and cultural factors as high-level factors. Among these, we find only societal factors contribute to an increase in algorithm acceptance, while others exacerbate algorithm aversion.

6. Research gaps and future research directions

We analyze the limitations acknowledged by the reviewed studies. Our review reveals a substantial acknowledgement related to methodological lacunae in algorithmic decision-making research. Further, we identify certain gaps and limitations in existing literature regarding research design and context. We also point out themes that remained unexplored. We use these gaps and limitations to suggest potential avenues for research.

6.1. Research design and generalizability

Several studies have acknowledged that their findings suffer from a lack of generalizability due to inherent limitations associated with research design (Araujo et al., 2020; Bigman and Gray, 2018; Dietvorst et al., 2015; Dijkstra, 1999; Eastwood et al., 2012; Laakasuo et al., 2021; Lennartz et al., 2021; Li et al., 2020; Önkal et al., 2019; Qiu and Benbasat, 2008; Sutherland et al., 2016; Swinney, 1999). The first of these limitations relates to sample selection, such as small sample (Kawaguchi, 2021; Lennartz et al., 2021; Swinney, 1999), unrepresentative sample such as only females (Laakasuo et al., 2021; Li et al., 2020), only students (Eastwood et al., 2012), or only professionals (Önkal et al., 2019), samples from certain geographic areas (Araujo et al., 2020; Bigman and Gray, 2018; Lennartz et al., 2021), and non-probability sampling selection (Laakasuo et al., 2021). As such, the internal and external validity of the findings may have been limited. Future researchers may address these limitations by undertaking more cross-cultural studies with representative samples (Kaur et al., 2021).

Second, our SLR finds that researchers use mostly scenario or vignette-based online and/or laboratory experiments, which may have limited ecological validity. In vignette-based experiments, subjects are told to imagine the situation and asked to respond accordingly. In the real world, however, the situation might be different (Goodwin et al., 2013; Shaffer et al., 2012); therefore, it is difficult to accurately mirror the real-world process in a vignette (Atzmüller and Steiner, 2010). The motivations for correct decisions and consequences of incorrect decisions in the real world are much more significant than for the decisions in online and laboratory experiments (Dietvorst et al., 2015; Yuviler-Gavish and Naseraldin, 2016). This difference elicits different levels of mental effort in judging the situation and making decisions. Furthermore, experiments based on vignettes rely on "subjects' perceptions rather than their actual behaviours" (Zhang et al., 2021) and what they want to do rather than what they did (Castelo et al., 2019; Thurman et al., 2019). Therefore, future research may be devoted to investigating algorithm aversion in a real-world setting (e.g., field experiment) that involves people's actual experience (Bigman and Gray, 2018; Lee, 2018; Yuviler-Gavish and Naseraldin, 2016).

Third, there is an increasing tendency among researchers to use crowd-workers and student communities for their experiments. Samples from crowd-workers have limitations. They tend to be younger (Prahl and Swol, 2021), liberal (Lee, 2018), and tech-savvy (Berger et al., 2021). They are more likely comfortable with technology and might have a greater propensity to follow algorithmic decisions (Bogert et al., 2021). Besides, due to a lack of real-world experience in decision-making, these groups lack sufficient acumen to reflect decisions that would be taken in the real world (Eastwood et al., 2012; Liu et al., 2019). As such, it is important to prioritize algorithm aversion research on other populations.

Fourth, we find that all reviewed studies use quantitative research methods. Though quantitative research is an effective way to establish and measure algorithm aversion, qualitative research may allow us to get in-depth insights about the underlying motives for algorithm aversion. Moving forward, scholars should undertake more qualitative research on this area, involving practitioners.

Fifth, the current study indicates that algorithm aversion research is mainly concentrated in highly developed countries. Research in the contexts of developing countries, on the other hand, may provide different insights. Different factors may account for aversion to algorithms in developing countries. For example, Gao et al. (2020) found that people in China are less concerned about the privacy risks associated with technology; thereby, they may display less aversion to algorithms. Similarly, Lennartz et al. (2021) believe that people from developing countries might exhibit a stronger preference for disruptive technology such as algorithmic decision. Moreover, due to the differences in institutional environments and resources (Yamakawa et al., 2008), the factors for aversion to algorithms in developing countries might significantly differ from those of developed countries. As such, future research may be directed to investigate the factors responsible for algorithm aversion in developing countries' contexts.

6.2. Theoretical framework and measurement

Our review signals a lack of a unified theoretical framework of algorithm aversion explaining its nature, antecedents, moderators, mediators, and outcomes. Absence of a theoretical framework makes it harder to obtain a holistic overview of algorithm aversion. The theoretical framework is necessary for understanding different concepts associated with it and guiding research (Duan et al., 2019). In our reviewed studies, we find that scholars mainly focused on merely one or few concepts related to algorithm aversion in their studies, for example, integration of people's opinions in algorithm design is explicated in Kawaguchi (2021), people's sensitivity to error in Dietvorst and Bharti (2020), and moral decisions in Niszczota and Kaszás (2020). Surprisingly, however, we do not have any comprehensive framework encompassing all related concepts.

Furthermore, extant literature does not suggest any scale for the algorithm aversion construct. Therefore, a unified scale of algorithm aversion is required. Such a scale will allow for statistical analysis to determine relationships among different variables of interest. As such, we concur and suggest the need for future research to develop a comprehensive theoretical framework as well as a measurement scale of the algorithm aversion construct.

6.3. Integrating a macro perspective

The current SLR study reveals that scholars' research foci have mostly centered around micro-level individual factors responsible for algorithm aversion. Therefore, firm factors are not explored adequately. However, algorithm utilization is largely influenced by different industry-specific factors (Gao and Sunyaev, 2019). We believe that algorithm adoption can also be viewed as a macro-level firm activity since many algorithmic decision-making tools are used by firms. We find only a few firm factors such as firms' nature (for-profit vs. not-for-profit) and role in sales channel (selling own product vs. consultancy) are discussed in the reviewed literature. Therefore, there is a lack of research that systematically examines the influence of different firm-level factors on the utilization of algorithmic decision. Only recently have we found few other firm-level factors such as facilitating conditions, performance and effort expectancy, perceived threat, etc. that affect managers' attitudes on using AI in organizational decision-making (Cao et al., 2021).

As such, future researchers might consider examining how algorithm adoption is influenced by different firm-level factors such as IT orientation (Zhou et al., 2005), dynamic capability (Teece et al., 1997), AI capability (Mikalef and Gupta, 2021), firms' resources (Barney, 2001), strategic orientation (Gatignon and Xuereb, 1997), risk sensitivity (Gönül et al., 2006), organizational readiness and AI strategy (Enholm et al., 2021), and manager's attitudes (Cao et al., 2021). While IT orientation pushes firms to acquire and apply new technology (Zhou et al., 2005), the dynamic capability allows firms to mobilize resources to cope with and seize the benefit of rapidly changing technological environment. Similarly, firms' AI capability can increase organizations' creativity and performance (Mikalef and Gupta, 2021). Consequently, it is also intriguing to see the magnitude of firms' performance as attributed to algorithm adoption. Future researchers might also consider investigating the impacts of firm-level factors that were explored in other organizational contexts such as manufacturing, supply chain, and product development on decision-making context. For example, Grover et al. (2020) identified six factors (job-fit, complexity, perceived consequences, affect towards use, social factors, and facilitating conditions) responsible for the use of AI in operation management context. Similarly, Enholm et al. (2021) found that different organizational factors such as culture, top management support, organizational readiness, employee-AI trust, AI strategy, and compatibility influence the use of AI in deriving business value. These factors may also influence the use of AI in decision-making context.

6.4. Exploring AI decision algorithm

Although few of our reviewed articles have discussed AI-based algorithmic decisions (Köbis and Mossink, 2021; Lennartz et al., 2021; Li et al., 2020; Sharan and Romano, 2020; van Esch et al., 2020), there is still a dearth of in-depth research on AI decisions. Many questions hitherto remain unanswered. For example, how can a newly formed firm build an AI-based algorithm when it lacks sufficient training data? Is historically constructed training data sufficiently ethical, correct, and efficient for future decisions? Is there any differential impact of a rule-based AI system and ML-based AI system on algorithmic decisions? How do people respond to these? Does perceived competitiveness of algorithms in respect to competitors' algorithms affect adoption? AI algorithm adoption may benefit from addressing these issues. A few research directions in this connection are adumbrated below.

First, newly established firms suffer from start-up problems due to a lack of data to build AI decision systems (Gogoll and Uhl, 2018). In this regard, future research may be undertaken to find out ways to overcome this problem. One possible research topic could be to investigate the possibility of collaboration for data sharing between peer firms (Bigdeli et al., 2013). This may enable new firms to gather similar types of data from their peer firms to feed into their system, which eventually can assist them to train the AI system. However, this needs to be done with the utmost caution, as there are risks of violation of data protection regulations, confidentiality, privacy, and security. Future research should also focus on designing frameworks or principles for this collaboration.

Second, the collaboration regarding data sharing may be extended to co-opetition (Brandenburger and Nalebuff, 1996), enabling participating firms to benefit mutually, creating an opportunity to access data from each other. This may empower the algorithms of participating firms to generate more reliable decisions based on big data (Khanra et al., 2020). Additionally, co-opetition may also help the partnering firms to handle unknown situations. For example, a situation that is unknown and new to one firm may not be new to other firm(s). In this instance, a firm that faces this new situation may benefit from collaborating with system(s) that have handled similar cases previously.

Third, Berger et al. (2021) posit that algorithm aversion could be overcome by demonstrating algorithms' ability to learn. The same is echoed by Dietvorst (Cockrell, 2020) when he posited that people may be more likely to believe that machine-learning would improve algorithms' performance and would have a "relatively higher chance of being near perfect." However, although machine-learning-based AI decisions may increase people's trust in algorithms, it may also spur some other firm-level challenges. Firms often operate in uncertain and competitive environments, in which they need to make decisions while considering competitors' decisions. We speculate that people's perception towards their own AI decision algorithms might be greatly influenced by the perception of the quality of competitors' AI decision algorithms. For example, imagine a firm "A," which has been established recently, has limited resources (e.g., data, computing power, skills) in comparison to its competitor "B." The algorithmic decisions generated by machine-learning algorithms of B might be more reliable and trustworthy than those of A, ceteris paribus. Given this situation, it is highly likely that firm A will avert its own algorithmic decisions, thinking them inferior to those of competitor B. As such, we believe that perceived competitiveness of algorithms in respect of competitors' algorithms may affect the adoption of algorithms and their decisions. Future research may be undertaken to test the validity of this intuition.

Fourth, researchers argue that algorithms perform better than human experts in decision-making tasks (Dietvorst et al., 2015). Nevertheless, extant literature has found that people prefer decisions from humans to algorithms (Dietvorst et al., 2015; Prahl and Swol, 2021). Future research must resolve this counterintuitive behavioral phenomenon. One possible way could be to generate decisions considering both experts' decisions and algorithms' decisions. Therefore, future research may be undertaken to see how to accommodate experts' decisions in algorithmic decision-making processes.

Fifth, AI decisions spur significant ethical challenges (Floridi and Taddeo, 2016). For example, based on an SLR on the use of AI in healthcare, Trocin et al. (2021) identified several ethical concerns such as transparency of data and their use, interpretability of how decisions are generated, unfair and biased decisions, unexpected profiling of the subject, and violation of privacy. Scholars could address these ethical concerns, responding to the call for research by Trocin et al. (2021). Research may also be conducted pursuing discourse ethics, which is grounded in actual discussion among those affected by decisions, to build responsible AI decisions (Mingers and Walsham, 2010).

Finally, in our SLR, we do not find any study that performs the comparative analysis on differential benefits of rule-based AI decision systems and machine-learning-based AI decision systems and their impacts on people's attitudes towards algorithmic decisions. Both systems have their own sets of advantages. A major advantage of a rule-based system is its ability to explain the reasoning of its decisions (Cao et al., 2021). In contrast, an ML-based system can learn from its data over time. Given these differential benefits, people may exhibit different levels of preference for rule-based systems and ML-based systems. Thus, we argue for the need to conduct comparative studies to advance the research area of AI decisions.

7. Discussion

7.1. Key findings

People tend to rely less on algorithms even when algorithms provide better decisions. Recently, this phenomenon has attracted many scholars to investigate why this happens and how it can be overcome. To summarize and interpret extant scholarship, we undertook an SLR following protocols suggested in previous literature (Kitchenham and Charters, 2007). Our primary objectives were to understand the current state-of-research profile, investigate the influencing factors of algorithm aversion (RQ1), and discover gaps and limitations to suggest future research directions (RQ2). We reviewed eighty empirical and peer-reviewed journal studies for this purpose.

Our findings indicate that most of the identified primary studies are exclusively experimental studies ($n = 69$) and nearly 50% of these studies are conducted within the last three years, indicating a growing interest in this topic. The subjects of the studies are mostly university students and crowdsource workers. This topic draws attention from a versatile group of research communities, including information technology, psychology, and business and management. Almost all studies are conducted in developed countries and data are analyzed quantitatively. However, some studies lack internal and external validity due to flawed sample selection and research design.

Our study identifies algorithm, individual, task, and high-level factors, which influence the acceptance of algorithmic decision-making. We further propose a comprehensive framework consisting of these factors and their relationship with algorithm aversion. This framework could guide future research on this subject. To advance algorithm aversion research, research gaps and limitations are highlighted, and several research directions are proposed. The findings of our SLR have important implications for researchers and practitioners.

7.2. Implications for research

This study provides several relevant contributions that could benefit researchers. First, the authors thoroughly reviewed extant empirical literature on algorithm aversion in a systematic manner and highlighted the understudied areas for scholarly attention. Based on the research profiling, implications arise for the need for research in real-world settings, applying qualitative research design and covering representative samples. This finding is aligned with the recommendation of Zhang et al. (2021), who assert that real-world studies are needed to calibrate the findings of experimental studies. Thus, the current study calls for designing algorithm aversion research to achieve more generalizable findings.

Second, the present study contributes theoretically by developing a framework for algorithm aversion. To do so, the authors synthesized previous works by summarizing key contributions of extant literature. Then, an inductive thematic analysis was performed to generate pertinent themes, maintaining a rigorous protocol. The proposed framework could be used as a groundwork for researchers to further explore algorithm aversion. The relationships depicted in the framework could be investigated and validated, utilizing different theoretical frameworks such as the technology acceptance model (TAM) (Davis, 1989), unified theory of acceptance and use of technology (UTAUT) (Venkatesh et al., 2003), technology threat avoidance theory (TTAT) (Liang and Xue, 2009), IS success model (DeLone and McLean, 2003), technology-organization-environment (TOE) framework (Tornatzky et al., 1990; Enholm et al., 2021), task-technology fit (TTF) (Goodhue, 1995), enablers-inhibitors perspective (Islam et al., 2020; Talukder et al., 2021), and person-technology fit (Tomer and Mishra, 2015). Utilizing these theories and models might also help to explore new factors responsible for and novel ways to overcome algorithm aversion. For example, integrating UTAUT and TTAT, Cao et al. (2021) developed the integrated AI acceptance-avoidance model (IAAAM), which helps to understand managers' attitudes and intentions towards using AI in decision-making.

Third, our study proposed a handful of future research directions with specific emphasis on AI algorithms, which draws relatively less attention from the research community. Notable among these are start-up problems, training experts' opinion, perception about competitors' algorithms, scale development of algorithm aversion, and consideration of organizational factors. We posit that data sharing collaboration and co-opetition could resolve start-up problems and help in increasing efficiency of algorithmic decisions. We propose that incorporating experts' opinion in algorithmic decision-making processes would increase the reliance on algorithms. We also speculate that perception about the quality of competitors' algorithms may shape people's attitudes regarding their algorithms. As such, we assume that testing these propositions will advance algorithm aversion research significantly.

Finally, this study also extends the findings of an existing SLR conducted by Burton et al. (2020) in the following ways. First, the study scope of Burton et al. (2020) is limited to augmented decision-making. Thus, it ignores the study of factors that are responsible for algorithm aversion in cases not related to augmented decision-making. In contrast, our study is broader and not restricted to any particular decision domain. Second, in contrast to Burton et al. (2020), our study uses an expansive set of search strings including algorithm appreciation, artificial intelligence, machine learning, and recommendations. We argue that due to not using an expansive set of search terms, their study may have missed many important papers related to algorithm aversion, particularly in the domain of AI. Third, there is a gap of over two years between running search terms by Burton et al. (2020) (i.e., December 2018) and us (i.e., March 2021). Therefore, our study is based on more recent publications. Fourth, the previous study uses only a database search, whereas we use both a database search and the snowballing technique for finding relevant papers. Fifth, we use a rigorous procedure and report the process in detail to ensure the replicability of the study.

We adopt an inductive approach to derive themes. On the other hand, the previous SLR uses a deductive approach by analyzing the studies on the basis of target variables. Sixth, we report 56 influencing factors of algorithm aversion under four major themes, many of which are not identified in the previous SLR. Seventh, we present a profile of the current state of the research of algorithm aversion, which the existing SLR lacks. Finally, we propose a conceptual framework delineating the relationships between the identified factors and algorithm aversion, which may serve as groundwork for future research. The existing SLR does not provide us any such model.

7.3. Implications for practitioners

Our study provides several practical implications for users and algorithm developers associated with algorithmic decision-making. First, developers of algorithmic decision-making should understand what factors cause aversion and what factors lead to adoption. Our current study presents a comprehensive list of several factors related to algorithm design, algorithmic decision, and delivery of algorithmic decision. This could guide developers in designing and developing algorithms. Developers can have a bird's eye view of the relevant determinants that need to be considered while planning for algorithm design. For example, it is evident that people reject algorithms due to its "black box"-type design. Therefore, there is a need for developers to consider unboxing the "black box" while developing algorithms. This also corroborates the earlier calls for research by several researchers (De Bruyn et al., 2020; Enholm et al., 2021; Trocin et al., 2021; Vlačić et al., 2021) who advocate for transparency and understandability of algorithmic decisions. Furthermore, our framework identifies several factors associated with individual characteristics, task nature, and other socio-environmental attributes, knowledge of which can benefit developers to understand for whom (individuals), for what (task), and for which conditions (high-level) algorithms need to be developed.

Second, the results of this study hold managerial implications regarding the use and implementation of the algorithmic decision system. Managers can develop deep insights from our findings about which factors while choosing or implementing algorithms need to be considered. For example, the study presents different task factors that indicate the domains of tasks, in which algorithmic decisions are suitable, and in which algorithms needs to be utilized with caution. Again, the study informs about different individual characteristics to management that will help managers to point out the factors responsible for resistance to or adoption of algorithms.

Third, our research points out different types of industries in which algorithmic decision-making has been examined so far (e.g., finance, healthcare, judiciary, sales, marketing, etc.). As such, our study serves as an important repository of knowledge derived from several experiments or use cases in those industries. Therefore, managers of these industries can use our framework to better understand algorithm aversion and formulate their strategic decisions.

8. Conclusion

Research indicates that for certain tasks, algorithms perform better than humans. Nevertheless, people show a greater degree of reluctance to follow algorithms. This behavioral anomaly attracts many researchers to investigate the phenomenon, resulting in a significant literature stream, each with its own explanation. Therefore, we feel the necessity to synthesize all prior findings and report the main themes. As such, we review 80 empirical studies to address four research questions. We attempt to answer those questions following both descriptive analysis and standard thematic analysis procedures. The analysis reveals that researchers investigated algorithm aversion using mostly vignette-based experimental research and that there is no unified scale to measure algorithm aversion in a real-world setting. Our SLR also finds that algorithm, individual, task, and high-level factors influence people's behavior associated with algorithm aversion. A framework delineating the relationships between these factors and algorithm aversion is presented for in-depth exploration in future. The study has identified some major lacunae in the existing literature and provided a set of directions, with a special emphasis on AI decision-making, to advance the research on algorithmic decision-making. Future research is solicited to develop theory and measurement scales. We believe that our findings would contribute both theoretically and practically to algorithmic decision-making.

However, despite multiple contributions of this study, it suffers from a few limitations that future research may take up. First, for our review, we only consider empirical studies appearing in English-language peer-reviewed journals. We restrict our search to a few selected academic databases; thus, some relevant studies may be missing from the current study. Future SLR studies can be conducted in a more comprehensive way including conference papers, book chapters, conceptual papers, and studies published in other languages. Second, we use a limited number of search terms. Future research may address this limitation by including a more expansive set of keywords. Third, we set a benchmark point for passing quality assessment criteria at fifty percent of the total achievable points. This benchmark might be too stringent. Future research may consider this benchmark liberally. Fourth, we consider several terms (e.g., robo-advisor, decision support system, and machine decision) analogous to an algorithm. We acknowledge that such consideration may have bias in the sample selection. Future research may find ways to overcome this subjective bias while performing an SLR on algorithm aversion.

Declaration of Competing Interest

None

Appendix A. Quality assessment score of primary studies

Sl. No.	Author(s)	QAC_1	QAC_2	QAC_3	QAC_4	QAC_5	*QS**
1	Kawaguchi (2021)	2	2	1.5	3	2	10.5
2	Köbis and Mossink (2021)	1	2	1.5	0	2	6.5
3	Berger et al. (2021)	2	2	3	3	2	12
4	Bogert et al. (2021)	2	2	1.5	0	2	7.5
5	Prahl and Swol (2021)	2	2	1.5	0	2	7.5
6	Fenneman et al. (2021)	2	2	1.5	0	2	7.5
7	Zhang et al. (2021)	2	2	3	3	2	12
8	Lennartz et al. (2021)	2	2	3	3	2	12
9	Laakasuo et al. (2021)	1	2	1.5	0	2	6.5
10	Yun et al. (2021)	2	2	1.5	0	2	7.5
11	Dietvorst and Bharti (2020)	2	2	3	3	2	12
12	Stein et al. (2020)	2	2	1.5	0	2	7.5

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Sl. No.	Author(s)	QAC_1	QAC_2	QAC_3	QAC_4	QAC_5	QS*
13	Niszczota and Kaszás (2020)	2	2	1.5	0	2	7.5
14	Litterscheidt and Streich (2020)	2	2	1.5	3	2	10.5
15	Feng and Gao (2020)	2	2	1.5	1.5	2	9
16	Efendić et al. (2020)	2	2	3	1.5	2	10.5
17	Araujo et al. (2020)	1	2	1.5	0	2	6.5
18	Sharan and Romano (2020)	2	2	1.5	1.5	2	9
19	Li et al. (2020)	1	2	0	1.5	2	6.5
20	Longoni and Cian (2020)	1	2	1.5	1.5	2	8
21	Toma et al. (2020)	1	2	1.5	0	2	6.5
22	van Esch et al. (2020)	1	2	1.5	0	2	6.5
23	Lourenço et al. (2020)	2	2	3	0	2	9
24	Önkal et al. (2019)	2	2	3	3	2	12
25	Castelo et al. (2019)	2	2	3	3	2	12
26	Yeomans et al. (2019)	2	2	3	3	2	12
27	Liu et al. (2019)	2	2	1.5	0	2	7.5
28	Thurman et al. (2019)	1	2	1.5	0	2	6.5
29	Logg et al. (2019)	2	2	3	1.5	2	10.5
30	Grgić-Hlača et al. (2019)	2	2	1.5	0	2	7.5
31	Park et al. (2019)	2	2	1.5	1.5	2	9
32	Pearson et al. (2019)	2	2	1.5	0	2	7.5
33	Luo et al. (2019)	1	2	1.5	1.5	2	8
34	Dietvorst et al. (2018)	2	2	1.5	3	2	10.5
35	Alexander et al. (2018)	2	2	0	1.5	2	7.5
36	Bigman and Gray (2018)	2	2	1.5	3	2	10.5
37	Lee (2018)	2	2	1.5	1.5	2	9
38	Gunaratne et al. (2018)	2	2	0	1.5	2	7.5
39	Gogoll and Uhl (2018)	2	2	1.5	0	2	7.5
40	Prahl and Van Swol (2017)	2	2	3	3	2	12
41	Goodyear et al. (2017)	2	2	1.5	0	2	7.5
42	Yuviler-Gavish and Naseraldin (2016)	2	2	1.5	0	2	7.5
43	Sutherland et al. (2016)	2	2	3	0	2	9
44	Dietvorst et al. (2015)	2	2	3	0	2	9
45	van Dongen and van Maanen (2013)	2	2	1.5	0	2	7.5
46	Goodwin et al. (2013)	2	2	1.5	0	2	7.5
47	McBride et al. (2012)	2	2	1.5	1.5	2	9
48	Pak et al. (2012)	2	2	1.5	0	2	7.5
49	Manzey et al. (2012)	2	2	1.5	0	2	7.5
50	Eastwood et al. (2012)	1	2	1.5	0	2	6.5
51	Shaffer et al. (2012)	2	2	1.5	0	1	6.5
52	Yuviler-Gavish and Gopher (2011)	2	2	0	1.5	2	7.5
53	Diab et al. (2011)	2	2	1.5	0	2	7.5
54	Kayande et al. (2009)	2	2	3	1.5	2	10.5
55	Önkal et al. (2009)	2	2	1.5	0	2	7.5
56	Rau et al. (2009)	2	2	3	0	2	9
57	Qiu and Benbasat (2008)	1	2	0	1.5	2	6.5
58	Merritt and Ilgen (2008)	2	2	1.5	0	2	7.5
59	Madhavan and Wiegmann (2007)	2	2	1.5	1.5	2	9
60	Hung et al. (2007)	2	2	1.5	0	2	7.5
61	Arkes et al. (2007)	2	2	1.5	0	2	7.5
62	Promberger and Baron (2006)	2	2	1.5	0	2	7.5
63	Gönül et al. (2006)	2	2	1.5	0	2	7.5
64	Madhavan et al. (2006)	2	2	1.5	0	2	7.5
65	Ho et al. (2005)	2	2	1.5	0	2	7.5
66	Workman (2005)	2	2	1.5	0	2	7.5
67	Sieck and Arkes (2005)	2	1	1.5	1.5	2	8
68	Fitzsimons and Lehmann (2004)	2	2	1.5	1.5	2	9
69	Sanders and Manrodt (2003)	1	2	1.5	0	2	6.5
70	Dzindolet et al. (2002)	1	2	1.5	1.5	2	8
71	Nass and Lee (2001)	1	2	1.5	0	2	6.5
72	Swinney (1999)	1	2	1.5	0	2	6.5
73	Dijkstra (1999)	2	2	1.5	0	2	7.5
74	Dijkstra et al. (1998)	2	2	1.5	0	2	7.5
75	Landsbergen et al. (1997)	2	2	1.5	1.5	2	9
76	Whitecotton (1996)	2	2	1.5	0	2	7.5
77	Lim and O'Connor (1996)	2	2	1.5	0	2	7.5
78	Mackay and Elam (1992)	2	2	1.5	0	2	7.5
79	Arkes et al. (1986)	2	2	1.5	0	2	7.5
80	Green and Hughes (1986)	2	2	1.5	1.5	2	9

* QS: Quality Score

Appendix B. Items for data extraction and their relevancy with research questions

Sl. No.	Data item	Description	Relevancy	Data category
1	Title	Title of the study		
2	Author(s)	Author(s) of the study		
3	Year	Publishing year of the study	Demographic	Contextual/ demographic
4	Journal name	Name of journal in which study was published	Demographic	Contextual/ demographic
5	Definition	Definition of the key terms (algorithm aversion/appreciation)	RQ1	Thematic
6	Objective of the study	Aim for which study was undertaken	QAC1	Protocol
7	Discussed decisions	Whether the study discusses "decision" component	IC1	Protocol
8	Interaction between human and algorithmic decisions	Whether there is any interaction between human and algorithmic decision	IC2	Protocol
9	Findings	Findings reported in the study with regards to current study	QAC4	Thematic
10	Causes of aversion	Reason(s) of algorithm aversion discussed in the study	QAC3, RQ1	Thematic
11	Overcoming aversion	Way of overcoming aversion outlined in the study	QAC4, RQ1	Thematic
12	Reasons for appreciation	Factors responsible for increasing appreciation	QAC3, RQ1	Thematic
13	Research method	Research method used in the study. For example, experiment, case study, etc.	QAC2	Contextual/ demographic
14	Data collection method	Methods used for collecting data. For example, survey, interview, experiment, etc.	QAC2	Contextual/ demographic
15	Data analysis type	How the collected data were analyzed and interpreted (qualitative, quantitative)	QAC2	Contextual/ demographic
16	Country	Country involved in the research	Contextual, RQ1	Contextual/ demographic
17	Decision domain	Decision area (industry, government, entertainment, etc.) examined	Contextual, RQ1	Contextual/ demographic
18	Decision type	Type of decision (business, moral etc.) discussed	Contextual, RQ1	Contextual/ demographic

Appendix C. Number of included studies by publisher and publication year

Publisher	# of Studies	Publication Year
Elsevier	24	2021, 2020(6), 2019(3), 2018(3), 2017, 2016, 2013(2), 2009, 2007, 2006, 2005, 2003, 1996, 1986
SAGE	11	2020(2), 2018, 2013, 2012, 2011, 2008, 2007(2), 2006, 2002
Wiley-Blackwell	9	2020, 2019(2), 2012, 2011, 2009, 2006, 2005, 1999
Taylor & Francis	8	2019, 2018, 2017, 2012, 2008, 1996, 1998, 1986
Institute for Operations Research and the Management Sciences	6	2021, 2019, 2018, 2009, 2004, 1992
Springer Nature	4	2021(2), 2020(2)
Association for Computing Machinery	3	2019(2), 2016
PLOS	2	2021, 2020
Oxford University Press	2	2005, 1997
APA	2	2015, 2001
American Accounting Association	1	1996
University of Central Florida	1	2021
Science Publishing Group	1	2020
Inderscience Publishers	1	2012
IGI Global Publishing	1	2020
American Marketing Association	1	2019
JMIR Publications	1	2021
Emerald	1	2021
Nature	1	2021

Appendix D. Overview of different attributes of research design and focus area

Sl. No.	Author(s)	Research method	No. of studies	Sample	Research focus	Term used to refer to algorithm	Themes derived
1	Kawaguchi (2021)	Field Study (I)	1	1500 (appx.) Japanese workers	*Av* & *Ap*	Algorithm	Expertise, risk and uncertainty, integration, general distrust, fear of change, affect heuristics
2	Köbis and Mossink (2021)	Experiment (I, B)	2	576 Prolific	*Av*	AI	Integration, fear of change, motivation
3	Berger et al. (2021)	Experiment (I, B)	1	452 MTurk	*Av* & *Ap*	Algorithm	Algorithm's ability to learn, negative perception, general distrust, affect heuristics
4	Bogert et al. (2021)	Experiment (I, B, W)	3	1044 MTurk	*Av* & *Ap*	Algorithm	Accuracy, subjectivity
5	Prahl and Swol (2021)	Experiment	2	689 MTurk, US	*Av*	Machine advisor	Emotional trust

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Sl. No.	Author(s)	Research method	No. of studies	Sample	Research focus	Term used to refer to algorithm	Themes derived
6	Fenneman et al. (2021)	Experiment (I, W)	1	119 MTurk, US	*Av* & Ap	Algorithm	Status quo, affect heuristics,
7	Zhang et al. (2021)	Experiment (I)	3	562, MTurk US	*Av*	Robo-advisor	Explainability, understandability, response speed, interface anthropomorphism, algorithm's role, emotional trust, awareness about algorithm's expertise, subjectivity, user's feedback, risk and uncertainty
8	Lennartz et al. (2021)	Survey	1	229, patients in Germany	*Av*	Artificial Intelligence	Ability to learn, emotional trust, risk and uncertainty
9	Laakasuo et al. (2021)	Experiment (I, B)	2	548, MTurk, Finland	*Av*	Robot	Mode of delivery
10	Yun et al. (2021)	Experiment (B, W)	2	372 MTurk	*Av* & Ap	Artificial intelligence	Understandability
11	Dietvorst and Bharti (2020)	Experiment (I, B)	9	4820 MTurk	*Av*	Algorithm	Perceived expectation, extraversion, risk and uncertainty
12	Stein et al. (2020)	Experiment (B)	1	134, students and employees, Germany	*Av*	Digital agent	Algorithm's complexity, interface anthropomorphism
13	Niszczota and Kaszás (2020)	Experiment (I, B, W)	5	3828 MTurk	*Av*	Robo-advisor	Moral task
14	Litterscheidt and Streich (2020)	Experiment (I)	1	205, Students and ORSEE, Germany	Ap	Robo-advisor	Black box nature, training
15	Feng and Gao (2020)	Experiment (B)	2	167, students, China	*Av* & Ap	Decision Support System	Affect heuristics, risk and uncertainty
16	Efendić et al. (2020)	Experiment (I, B, W)	7	1928 MTurk, US	*Av*	Algorithm	Response speed
17	Araujo et al. (2020)	Experiment (B, W)	1	958, Kantar, Netherland	*Av* & Ap	Automated decision-making	Online self-efficacy, privacy concern, training, age, gender
18	Sharan and Romano (2020)	Experiment (B)	1	171, students, UK	*Av* & Ap	AI-based algorithm	Calibration feedback, self-efficacy, black box nature, neuroticism, familiarity with human advisor
19	Li et al. (2020)	Experiment (B, W)	2	70, China	*Av*	AI Recommendation	Interface anthropomorphism, perceived expectation, affect heuristics, training
20	Longoni and Cian (2020)	Experiment and field study (I, B)	7 (6 + 1)	2894 (experiment: 2465, field: 429) students and MTurk, US, Italy	Ap	AI	Subjectivity
21	Toma et al. (2020)	Experiment (B, W)	2	137 (MTurk and LinkedIn)	*Av* & Ap	Algorithmic aid, automated advice	Outlook (positive vs. negative)
22	van Esch et al. (2020)	Survey	1	532, panel US	*Av* & Ap	AI	Neuroticism, motivation
23	Lourenço et al. (2020)	Experiment and survey	2 (1 + 1)	1850 (experiment: 201; survey: 1649) SSI, Netherland	*Av* & Ap	Robo-Advice	Age, profit orientation, organization's type
24	Önkal et al. (2019)	Survey	1	134 Executive MBA students, Turkey	*Av* & Ap	Algorithm	Accessibility, explainability, understandability, self-efficacy, training
25	Castelo et al. (2019)	Experiment and field study (I, B)	6 (4 + 2)	57,400 (experiments: 1400; field studies: 56,000), MTurk, Prolific, Facebook	*Av* & Ap	Algorithm	Interface anthropomorphism, subjectivity
26	Yeomans et al. (2019)	Experiment (I, B, W)	5	3647 MTurk US	*Av*	Recommender system	Understandability, explanability
27	Liu et al. (2019)	Experiment (B)	5	1267 students, China	*Av*	Self-driving vehicle	Affect heuristics
28	Thurman et al. (2019)	Survey	1	53,314, audience of news, multiple countries	*Av* & Ap	Algorithm	Privacy concern, age, gender, education, familiarity with human advisor
29	Logg et al. (2019)	Experiment (I, B)	6	671 MTurk, academic researchers	*Av* & Ap	Algorithm	Self-esteem, self-efficacy, age, education
30	Grgić-Hlača et al. (2019)	Experiment (I, W)	3	103 Prolific, US	*Av*	Machine advice	Risk and uncertainty
31	Park et al. (2019)	Experiment (I, B)	3	372 students and MTurk, US	Ap	Algorithm	Response speed
32	Pearson et al. (2019)	Experiment (I, B, W)	1	200 MTurk, US	Ap	Automated adviser	Awareness about algorithms' expertise
33	Luo et al. (2019)	Field study (B)	1	6255 customers, Asia	*Av*	Chatbot	Negative perception, training
34	Dietvorst et al. (2018)	Experiment (I, B)	3	1922 students and MTurk, US	*Av*	Algorithm	Calibration feedback, locus of control
35	Alexander et al. (2018)	Experiment (I)	1	46 students, US	Ap	Algorithm	User's feedback
36	Bigman and Gray (2018)	Experiment (I, B, W)	9	2473 MTurk, US and Canada	*Av*	Machine (making decision)	Algorithms' role, awareness about algorithms' expertise, moral task
37	Lee (2018)	Experiment (I, B, W)	9	228 MTurk, US	*Av*	Algorithm (Mechanical skills)	Self-esteem, subjectivity, moral task
38	Gunaratne et al. (2018)	Experiment (I, B)	1	314 MTurk	Ap	Algorithmic decision	Awareness about algorithms' expertise
39	Gogoll and Uhl (2018)	Experiment (I)	3	264 ORSEE, Germany	*Av*	Machine (making decision)	Emotional trust, moral task

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Sl. No.	Author(s)	Research method	No. of studies	Sample	Research focus	Term used to refer to algorithm	Themes derived
40	Prahl and Van Swol (2017)	Experiment (I)	1	157 students, US	AV	Algorithm	General aversion, fear of change, affect heuristics, length of familiarity
41	Goodyear et al. (2017)	Experiment (I, B, W)	2	47 students, US	*Av*	Machine agents	Preconceived expectation
42	Yuviler-Gavish and Naseraldin (2016)	Experiment (I, B, W)	3	100 students, Israel	*Av*	Decision aid	Affect heuristics
43	Sutherland et al. (2016)	Experiment (B, W)	1	145 students, US	*Av* & Ap	Automated	Cost of advice, egocentric trimming, self-esteem, awareness about algorithms' expertise, risk and uncertainty
44	Dietvorst et al. (2015)	Experiment (I)	5	921 students and MTurk, US	*Av*	Algorithm	Feedback only, ability to learn, affect heuristics
45	van Dongen and van Maanen (2013)	Experiment (I)	2	79 students, Netherland	*Av* & Ap	Decision aid	Calibration feedback, feeling about responsibility, serial position effect
46	Goodwin et al. (2013)	Experiment (I, B, W)	2	92 students, Turkey	*Av* & Ap	Forecasting support systems (FSSs)	Accessibility, explainability, integrating
47	McBride et al. (2012)	Experiment	2	55 nurses, US	*Av* & Ap	Automated decision aids (ADAs)	Personality traits (judger vs. perceiver)
48	Pak et al. (2012)	Experiment (I, B)	1	90 students and volunteers, US	*Av* & Ap	Decision aid	Interface anthropomorphism
49	Manzey et al. (2012)	Experiment (I, B, W)	2	144 students, Germany	*Av*	Automated decision aid	Serial position effect (algorithms' error)
50	Eastwood et al. (2012)	Experiment (I, B, W)	3	235 students, Canada	*Av*	Actuarial decision aids	Other people's views
51	Shaffer et al. (2012)	Experiment	3	434 students, US	*Av*	Computerized clinical decision support systems (CDSSs)	Negative perception, locus of control, relationship with human advisor
52	Yuviler-Gavish and Gopher (2011)	Experiment (I, B, W)	3	39 students, Israel	Ap	Decision support systems for the long term (DSSLT)	Serial position effect (algorithm introduction)
53	Diab et al. (2011)	Experiment (B)	1	418 Study Response Project, US	*Av*	Decision aid	Other people's views
54	Kayande et al. (2009)	Experiment (I)	1	81 students, US	*Av*	Decision Support System	Accessibility, outlook
55	Önkal et al. (2009)	Experiment (I)	2	130 students, Turkey	*Av*	Statistical method	Accessibility, mode of delivery, general aversion, complexity
56	Rau et al. (2009)	Experiment	1	16 students, Germany and China	*Av* & Ap	Robot	Culture: individualistic vs. collectivistic
57	Qiu and Benbasat (2008)	Experiment (I)	1	168 students and employees, US	Ap	Robot	Interface anthropomorphism
58	Merritt and Ilgen (2008)	Experiment (I, W)	1	255 students, US	*Av* & Ap	Automation	Extraversion, affect heuristics, frequency of use
59	Madhavan and Wiegmann (2007)	Experiment (I, B, W)	2	222 students, US	*Av*	Decision aid	Interface anthropomorphism, complexity, preconceived expectations
60	Hung et al. (2007)	Experiment (I)	1	65 students, Taiwan	Ap	Decision Support System	Affect heuristics
61	Arkes et al. (2007)	Experiment (I)	4	680 students, patients, US	*Av*	Decision Support System	Other people's view
62	Promberger and Baron (2006)	Experiment (I, W)	2	166 online, US	*Av*	Decision from computer program	Emotional trust, perceived responsibility
63	Gönül et al. (2006)	Experiment (I)	1	116 students, Turkey	*Av* & Ap	Decision Support System	Explainability, structural characteristics of delivery
64	Madhavan et al. (2006)	Experiment (I)	1	45 students, US	*Av*	Automation	Accuracy
65	Ho et al. (2005)	Experiment (I)	2	49 community members, Canada	*Av* & Ap	Medication management system	Age
66	Workman (2005)	Survey	1	209 engineers, US	*Av* & Ap	Expert system decision support (EDSS)	Gender, other people's view
67	Sieck and Arkes (2005)	Experiment (I, B)	3	337 students and employee, US	*Av*	Decision aid	Calibration feedback, self-efficacy
68	Fitzsimons and Lehmann (2004)	Experiment (I, B)	4	438 students, US	*Av*	Decision Support System	Integration
69	Sanders and Manrodt (2003)	Survey	1	240 firms, US	Ap	Quantitative method	Organization's type
70	Dzindolet et al. (2002)	Experiment (I, B)	3	273 students, US	*Av*	Automation aid	Preconceived expectation, moral obligation to follow own decision, black box nature, locus of control, training
71	Nass and Lee (2001)	Experiment (I, B)	2	152 students, US	*Av* & Ap	Computer-synthesized speech	Integration
72	Swinney (1999)	Experiment	1	29 auditors, US	Ap	Expert system	Outlook of decision
73	Dijkstra (1999)	Experiment (I)	2	73 students, Netherland	Ap	Expert system	Motivation, serial position effect (algorithm introduction)
74	Dijkstra et al. (1998)	Experiment (I)	1	85 students, Netherland	Ap	Expert system	Subjectivity
75		Experiment	1	101 students, US	*Av*	Expert system	Integration, self-efficacy

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Sl. No.	Author(s)	Research method	No. of studies	Sample	Research focus	Term used to refer to algorithm	Themes derived
	Landsbergen et al. (1997)						
76	Whitecotton (1996)	Experiment	1	75 (35 students and 40 professional financial analysts), US	*Av*	Decision aid	Self-efficacy, status quo, affect heuristics, familiarity with task
77	Lim and O'Connor (1996)	Experiment (I, B, W)	2	64 students and employees, Australia	*Av*	Decision Support System	Motivation, familiarity with algorithm
78	Mackay and Elam (1992)	Experiment (I)	1	12 US	*Av*	Decision aid	Awareness about algorithms' expertise
79	Arkes et al. (1986)	Experiment (I)	2	226 students, US	*Av*	Decision aid	Motivation
80	Green and Hughes (1986)	Experiment (B, W)	1	63 managers, US	*Av*	Decision Support System	Training

Note: I: Incentivize, B: Between-subjects, W: Within-subjects, Av: Aversion, Ap: Appreciation

Reference

Acharya, A., Singh, S.K., Pereira, V., Singh, P., 2018. Big data, knowledge co-creation and decision making in fashion industry. Int. J. Inf. Manage. 42, 90–101. https://doi.org/10.1016/J.IJINFOMGT.2018.06.008.

Alexander, V., Blinder, C., Zak, P.J., 2018. Why trust an algorithm? Performance, cognition, and neurophysiology. Comput. Human Behav. 89, 279–288. https://doi.org/10.1016/J.CHB.2018.07.026.

Araujo, T., Helberger, N., Kruikemeier, S., de Vreese, C.H., 2020. In AI we trust? Perceptions about automated decision-making by artificial intelligence. AI Soc. 35, 611–623. https://doi.org/10.1007/s00146-019-00931-w.

Arkes, H.R., Dawes, R.M., Christensen, C., 1986. Factors influencing the use of a decision rule in a probabilistic task. Organ. Behav. Hum. Decis. Process. 37, 93–110. https://doi.org/10.1016/0749-5978(86)90046-4.

Arkes, H.R., Shaffer, V.A., Medow, M.A., 2007. Patients derogate physicians who use a computer-assisted diagnostic aid. Med. Decis. Making 27, 189–202. https://doi.org/10.1177/0272989X06297391.

Atzmüller, C., Steiner, P.M., 2010. Experimental vignette studies n survey research. Methodology 6, 128–138. https://doi.org/10.1027/1614-2241/a000014.

Bandura, A., 2010. Self-Efficacy. The Corsini Encyclopedia of Psychology. John Wiley & Sons, Inc., Hoboken, NJ, USA https://doi.org/10.1002/9780470479216.corpsy0836.

Banijamali, A., Pakanen, O.P., Kuvaja, P., Oivo, M., 2020. Software architectures of the convergence of cloud computing and the Internet of Things: a systematic literature review. Inf. Softw. Technol. 122, 106271 https://doi.org/10.1016/J.INFSOF.2020.106271.

Bano, M., Zowghi, D., 2015. A systematic review on the relationship between user involvement and system success. Inf. Softw. Technol. 58, 148–169. https://doi.org/10.1016/J.INFSOF.2014.06.011.

Barbosa, A., Hovakimyan, N., Martin, N.F., 2020. Risk-averse optimization of crop inputs using a deep ensemble of convolutional neural networks. Comput. Electron. Agric. 178, 105785 https://doi.org/10.1016/j.compag.2020.105785.

Barney, J.B., 2001. Resource-based theories of competitive advantage: a ten-year retrospective on the resource-based view. J. Manage. 27, 643–650. https://doi.org/10.1177/014920630102700602.

Behera, R.K., Bala, P.K., Dhir, A., 2019. The emerging role of cognitive computing in healthcare: a systematic literature review. Int. J. Med. Inform. 129, 154–166. https://doi.org/10.1016/J.IJMEDINF.2019.04.024.

Behutiye, W.N., Rodríguez, P., Oivo, M., Tosun, A., 2017. Analyzing the concept of technical debt in the context of agile software development: a systematic literature review. Inf. Softw. Technol. 82, 139–158. https://doi.org/10.1016/J.INFSOF.2016.10.004.

Berger, B., Adam, M., Rühr, A., Benlian, A., 2021. Watch Me Improve—Algorithm Aversion and Demonstrating the Ability to Learn. Bus. Inf. Syst. Eng. 63, 55–68. https://doi.org/10.1007/s12599-020-00678-5.

Bhimani, H., Mention, A.L., Barlatier, P.J., 2019. Social media and innovation: a systematic literature review and future research directions. Technol. Forecast. Soc. Change 144, 251–269. https://doi.org/10.1016/J.TECHFORE.2018.10.007.

Bigdeli, A.Z., Kamal, M.M., de Cesare, S., 2013. Electronic information sharing in local government authorities: factors influencing the decision-making process. Int. J. Inf. Manage. 33, 816–830. https://doi.org/10.1016/J.IJINFOMGT.2013.05.008.

Bigman, Y.E., Gray, K., 2018. People are averse to machines making moral decisions. Cognition 181, 21–34. https://doi.org/10.1016/J.COGNITION.2018.08.003.

Bogert, E., Schecter, A., Watson, R.T., 2021. Humans rely more on algorithms than social influence as a task becomes more difficult. Sci. Rep. 11, 1–9. https://doi.org/10.1038/s41598-021-87480-9.

Brandenburger, A.M., Nalebuff, B.J., 1996. Co-Opetition. Doubleday, New York.

Braun, V., Clarke, V., 2006. Using thematic analysis in psychology. Qual. Res. Psychol. 3, 77–101. https://doi.org/10.1191/1478088706qp063oa.

Brehm, J.W., 1966. A Theory of Psychological Reactance. Academic Press, New York.

BSSR, 2019. BSSR Definition – Office of Behavioral and Social Sciences Research [WWW Document]. National Institute of Health. URL https://obssr.od.nih.gov/about/bssr-definition/#ref1 (accessed 7.2.21).

Burton, J.W., Stein, M.K., Jensen, T.B., 2020. A systematic review of algorithm aversion in augmented decision making. J. Behav. Decis. Mak. 33, 220–239. https://doi.org/10.1002/bdm.2155.

Cao, G., Duan, Y., Edwards, J.S., Dwivedi, Y.K., 2021. Understanding managers' attitudes and behavioral intentions towards using artificial intelligence for organizational decision-making. Technovation 106, 102312. https://doi.org/10.1016/J.TECHNOVATION.2021.102312.

Castelluccia, C., Le Métayer, D., 2019. Understanding algorithmic decision-making: opportunities and challenges. https://doi.org/10.2861/536131.

Castelo, N., Bos, M.W., Lehmann, D.R., 2019. Task-Dependent Algorithm Aversion. J. Market. Res. 56, 809–825. https://doi.org/10.1177/0022243719851788.

Chander, A., Wang, J., Srinivasan, R., Uchino, K., Chelian, S., 2018. Working with Beliefs: AI Transparency in the Enterprise, in: IUI Workshops.

Cheng, J., Zhu, L., Ke, Y., Chu, S., 2012. Fast algorithms for maximal clique enumeration with limited memory. In: Proceedings of the ACM SIGKDD International Conference on Knowledge Discovery and Data Mining, pp. 1240–1248. https://doi.org/10.1145/2339530.2339724.

Cheyne, H., Hundley, V., Dowding, D., Bland, J.M., McNamee, P., Greer, I., Styles, M., Barnett, C.A., Scotland, G., Niven, C., 2008. Effects of algorithm for diagnosis of active labour: cluster randomised trial. BMJ 337, 1396–1400. https://doi.org/10.1136/bmj.a2396.

Cockrell, J., 2020. Even when algorithms outperform humans, people often reject them [WWW Document]. Chicago Booth Rev. URL https://review.chicagobooth.edu/behavioral-science/2020/article/even-when-algorithms-outperform-humans-people-often-reject-them (accessed 6.26.21).

Collins, C., Dennehy, D., Conboy, K., Mikalef, P., 2021. Artificial intelligence in information systems research: a systematic literature review and research agenda. Int. J. Inf. Manage. 60, 102383 https://doi.org/10.1016/J.IJINFOMGT.2021.102383.

Colman, A.M., 2015. A Dictionary of Psychology, A Dictionary of Psychology. Oxford University Press. https://doi.org/10.1093/acref/9780199657681.001.0001.

Daming, L., Qinglang, S., Lianbing, D., Kaicheng, C., Zhiming, C., Mohammed, B.O., 2020. Load balancing mechanism in the cloud environment using preference alignments and an optimisation algorithm. ET Communications 14, 489–496. https://doi.org/10.1049/iet-com.2019.0800.

Davis, F.D., 1989. Perceived usefulness, perceived ease of use, and user acceptance of information technology. MIS Quarterly: Manage. Inf. Syst. 13, 319–339. https://doi.org/10.2307/249008.

De Bruyn, A., Viswanathan, V., Beh, Y.S., Brock, J.K.U., von Wangenheim, F., 2020. Artificial intelligence and marketing: pitfalls and opportunities. J. Interact. Market. 51, 91–105. https://doi.org/10.1016/j.intmar.2020.04.007.

DeLone, W.H., McLean, E.R., 2003. The DeLone and McLean model of information systems success: a ten-year update. J. Manage. Inf. Syst. 9–30. https://doi.org/10.1080/07421222.2003.11045748. M.E. Sharpe Inc.

Diab, D.L., Pui, S.Y., Yankelevich, M., Highhouse, S., 2011. Lay perceptions of selection decision aids in US and non-US samples. Int. J. Sel. Assess. 19, 209–216. https://doi.org/10.1111/J.1468-2389.2011.00548.X.

Dietvorst, B.J., Bharti, S., 2020. People Reject Algorithms in Uncertain Decision Domains Because They Have Diminishing Sensitivity to Forecasting Error. Psychol. Sci. 31, 1302–1314. https://doi.org/10.1177/0956797620948841.

Dietvorst, B.J., Simmons, J.P., Massey, C., 2018. Overcoming algorithm aversion: people will use imperfect algorithms if they can (even slightly) modify them. Manage. Sci. 64, 1155–1170. https://doi.org/10.1287/mnsc.2016.2643.

Dietvorst, B.J., Simmons, J.P., Massey, C., 2015. Algorithm aversion: people erroneously avoid algorithms after seeing them err. J. Exp. Psychol.: General 144, 114–126. https://doi.org/10.1037/xge0000033.

Dijkstra, J.J., 1999. User agreement with incorrect expert system advice. Behav. Inf. Technol. 18, 399–411. https://doi.org/10.1080/014492999118832.

Dijkstra, J.J., Liebrand, W.B.G., Timminga, E., 1998. Persuasiveness of expert systems. Behav. Inf. Technol. 17, 155–163. https://doi.org/10.1080/014492998119526.

Dikert, K., Paasivaara, M., Lassenius, C., 2016. Challenges and success factors for large-scale agile transformations: a systematic literature review. J. Syst. Softw. 119, 87–108. https://doi.org/10.1016/J.JSS.2016.06.013.
Duan, Y., Edwards, J.S., Dwivedi, Y.K., 2019. Artificial intelligence for decision making in the era of Big Data – evolution, challenges and research agenda. Int. J. Inf. Manage. 48, 63–71. https://doi.org/10.1016/J.IJINFOMGT.2019.01.021.
Dzindolet, M.T., Pierce, L.G., Beck, H.P., Dawe, L.A., 2002. The perceived utility of human and automated aids in a visual detection task. Hum. Factors 44, 79–94. https://doi.org/10.1518/0018720024494856.
Dzindolet, M.T., Pierce, L.G., Beck, H.P., Dawe, L.A., Anderson, B.W., 2001. Predicting Misuse and Disuse of Combat Identification Systems. Military Psychol. 13, 147–164. https://doi.org/10.1207/S15327876MP1303_2.
Eastwood, J., Snook, B., Luther, K., 2012. What people want from their professionals: attitudes toward decision-making strategies. J. Behav. Decis. Mak. 25, 458–468. https://doi.org/10.1002/bdm.741.
Ebbinghaus, H., 1913. Memory; a Contribution to Experimental Psychology. Teachers college, Columbia university, New York.
Efendić, E., van de Calseyde, P., van de Calseyde, P.P.F.M., Evans, A.M., 2020. Slow response times undermine trust in algorithmic (but not human) predictions. Organ. Behav. Hum. Decis. Process. 157, 103–114. https://doi.org/10.1016/j.obhdp.2020.01.008.
Elkins, A.C., Dunbar, N.E., Adame, B., Nunamaker, J.F., 2014. Are users threatened by credibility assessment systems? J. Manage. Inf. Syst. 29, 249–262. https://doi.org/10.2753/MIS0742-1222290409.
Enholm, I.M., Papagiannidis, E., Mikalef, P., Krogstie, J., 2021. Artificial intelligence and business value: a literature review. Inf. Syst. Front. https://doi.org/10.1007/S10796-021-10186-W.
Feng, C., Dong, Y., Jiang, Y., Ran, M., 2018. Dynamic search space particle swarm optimization approach for portfolio optimization. In: ACM International Conference Proceeding Series. Association for Computing Machinery, pp. 127–131. https://doi.org/10.1145/3232651.3232666.
Feng, X., Gao, J., 2020. Is optimal recommendation the best? A laboratory investigation under the newsvendor problem. Decis. Support Syst. 131, 113251 https://doi.org/10.1016/J.DSS.2020.113251.
Fenneman, A., Sickmann, J., Pitz, T., Sanfey, A.G., 2021. Two distinct and separable processes underlie individual differences in algorithm adherence: differences in predictions and differences in trust thresholds. PLoS One 16. https://doi.org/10.1371/JOURNAL.PONE.0247084.
Festinger, L., 1957. A Theory of Cognitive Dissonance. Stanford University Press.
Filiz, I., René Judek, J., Lorenz, M., Spiwoks, M., 2021. The Tragedy of Algorithm Aversion.
Fitzsimons, G.J., Lehmann, D.R., 2004. Reactance to Recommendations: when Unsolicited Advice Yields Contrary Responses. Market. Sci. 23, 82–94. https://doi.org/10.1287/MKSC.1030.0033.
Floridi, L., Taddeo, M., 2016. What is data ethics? Philos. Trans. R. Soc., A 374. https://doi.org/10.1098/RSTA.2016.0360.
Gao, F., Sunyaev, A., 2019. Context matters: a review of the determinant factors in the decision to adopt cloud computing in healthcare. Int. J. Inf. Manage. 48, 120–138. https://doi.org/10.1016/J.IJINFOMGT.2019.02.002.
Gao, S., He, L., Chen, Y., Li, D., Lai, K., 2020. Public perception of artificial intelligence in medical care: content analysis of social media. J. Med. Internet Res. 22, e16649. https://doi.org/10.2196/16649.
Garousi, V., Petersen, K., Ozkan, B., 2016. Challenges and best practices in industry-academia collaborations in software engineering: a systematic literature review. Inf. Softw. Technol. 79, 106–127. https://doi.org/10.1016/J.INFSOF.2016.07.006.
Gatignon, H., Xuereb, J.M., 1997. Strategic orientation of the firm and new product performance. J. Market. Res. 34, 77–90. https://doi.org/10.2307/3152066.
Genc-Nayebi, N., Abran, A., 2017. A systematic literature review: opinion mining studies from mobile app store user reviews. J. Syst. Softw. 125, 207–219. https://doi.org/10.1016/J.JSS.2016.11.027.
Gillespie, T., 2014. The Relevance of Algorithms. Media Technol. : Essays Commun. Mater. Soc. 167–194. https://doi.org/10.7551/mitpress/9780262525374.003.0009.
Gino, F., 2008. Do we listen to advice just because we paid for it? The impact of advice cost on its use Organ. Behav. Hum. Decis. Process. 107, 234–245. https://doi.org/10.1016/j.obhdp.2008.03.001.
Gogoll, J., Uhl, M., 2018. Rage against the machine: automation in the moral domain. J. Behav. Exp. Econ. 74, 97–103. https://doi.org/10.1016/J.SOCEC.2018.04.003.
Gönül, M.S., Önkal, D., Lawrence, M., 2006. The effects of structural characteristics of explanations on use of a DSS. Decis. Support Syst. 42, 1481–1493. https://doi.org/10.1016/J.DSS.2005.12.003.
Goodhue, D.L., 1995. Understanding User Evaluations of Information Systems. Manage. Sci. 41, 1827–1844. https://doi.org/10.1287/mnsc.41.12.1827.
Goodwin, P., Sinan Gönül, M., Önkal, D., 2013. Antecedents and effects of trust in forecasting advice. Int. J. Forecast. 29, 354–366. https://doi.org/10.1016/J.IJFORECAST.2012.08.001.
Goodyear, K., Parasuraman, R., Chernyak, S., de Visser, E., Madhavan, P., Deshpande, G., Krueger, F., 2017. An fMRI and effective connectivity study investigating miss errors during advice utilization from human and machine agents. Soc. Neurosci. 12, 570–581. https://doi.org/10.1080/17470919.2016.1205131.
Green, G.I., Hughes, C.T., 1986. Effects of decision support systems training and cognitive style on decision process attributes. J. Manage. Inf. Syst. 3, 83–93. https://doi.org/10.1080/07421222.1986.11517764.
Grgić-Hlača, N., Engel, C., Gummadi, K.P., 2019. Human decision making with machine assistance. Proc. ACM Human-Comput. Interaction 3. https://doi.org/10.1145/3359280.
Grover, P., Kar, A.K., Dwivedi, Y.K., 2020. Understanding artificial intelligence adoption in operations management: insights from the review of academic literature and social media discussions. Ann. Oper. Res. 2020, 1–37. https://doi.org/10.1007/S10479-020-03683-9.
Guckenbiehl, P., Corral de Zubielqui, G., Lindsay, N., 2021. Knowledge and innovation in start-up ventures: a systematic literature review and research agenda. Technol. Forecast. Soc. Change 172, 121026. https://doi.org/10.1016/J.TECHFORE.2021.121026.
Gunaratne, J., Zalmanson, L., Nov, O., 2018. The persuasive power of algorithmic and crowdsourced advice. J. Manage. Inf. Syst. 35, 1092–1120. https://doi.org/10.1080/07421222.2018.1523534.
Ho, G., Wheatley, D., Scialfa, C.T., 2005. Age differences in trust and reliance of a medication management system. Interact. Comput. 17, 690–710. https://doi.org/10.1016/J.INTCOM.2005.09.007.
Hu, Y.H., 1992. The Quantization Effects of the CORDIC Algorithm. IEEE Trans. Signal Process. 40, 834–844. https://doi.org/10.1109/78.127956.
Huang, M.-.H., Rust, R.T., 2020. A strategic framework for artificial intelligence in marketing. J. Acad. Market. Sci. 1 (49), 30–50. https://doi.org/10.1007/S11747-020-00749-9, 2020 49.
Hung, S.Y., Ku, Y.C., Liang, T.P., Lee, C.J., 2007. Regret avoidance as a measure of DSS success: an exploratory study. Decis. Support Syst. 42, 2093–2106. https://doi.org/10.1016/J.DSS.2006.05.006.
Islam, A.K.M.N., Cenfetelli, R., Benbasat, I., 2020. Organizational buyers' assimilation of B2B platforms: effects of IT-enabled service functionality. J. Strategic Inf. Syst. 29, 101597 https://doi.org/10.1016/J.JSIS.2020.101597.
Jones, M.v., Coviello, N., Tang, Y.K., 2011. International Entrepreneurship research (1989-2009): a domain ontology and thematic analysis. J. Bus. Ventur. 26, 632–659. https://doi.org/10.1016/j.jbusvent.2011.04.001.
Judge, T.A., Locke, E., Durham, C., 1997. The dispositional causes of job satisfaction: a core evaluations approach. Res. Organ. Behav. 19–151.
Judge, T.A., Locke, E.A., Durham, C.C., Kluger, A.N., 1998. Dispositional effects on job and life satisfaction: the role of core evaluations. J. Appl. Psychol. 17–34.
Jussupow, E., Benbasat, I., Heinzl, A., 2020. Why are we averse towards Algorithms? a comprehensive literature review on algorithm aversion, in: 28th European Conference on Information Systems (ECIS).
Kaur, P., Dhir, A., Tandon, A., Alzeiby, E.A., Abohassan, A.A., 2021. A systematic literature review on cyberstalking. an analysis of past achievements and future promises. Technol. Forecast. Soc. Change 163. https://doi.org/10.1016/j.techfore.2020.120426.
Kawaguchi, K., 2021. When Will Workers Follow an Algorithm? A Field Experiment with a Retail Business. Manage. Sci. 67, 1670–1695. https://doi.org/10.1287/mnsc.2020.3599.
Kayande, U., de Bruyn, A., Lilien, G.L., Rangaswamy, A., van Bruggen, G.H., 2009. How incorporating feedback mechanisms in a DSS affects DSS evaluations. Inf. Syst. Res. 20, 527–546. https://doi.org/10.1287/ISRE.1080.0198.
Khan, A., Krishnan, S., Dhir, A., 2021. Electronic government and corruption: systematic literature review, framework, and agenda for future research. Technol. Forecast. Soc. Change 167, 120737. https://doi.org/10.1016/j.techfore.2021.120737.
Khanra, S., Dhir, A., M Najmul Islam, A.K., Mäntymäki, M., 2020. Enterprise Information Systems Big data analytics in healthcare: a systematic literature review Big data analytics in healthcare: a systematic literature review. https://doi.org/10.1080/17517575.2020.1812005.
Kitchenham, B., Charters, S., 2007. Guidelines for performing systematic literature reviews in software engineering, Technical report, Ver. 2.3 EBSE Technical Report. EBSE.
Köbis, N., Mossink, L.D., 2021. Artificial intelligence versus Maya Angelou: experimental evidence that people cannot differentiate AI-generated from human-written poetry. Comput. Human Behav. 114, 106553 https://doi.org/10.1016/J.CHB.2020.106553.
Kolbinger, O., Knopp, M., 2020. Video kills the sentiment—Exploring fans' reception of the video assistant referee in the English premier league using Twitter data. PLoS One 15, e0242728. https://doi.org/10.1371/journal.pone.0242728.
Laakasuo, M., Palomäki, J., Köbis, N., 2021. Moral Uncanny Valley: a Robot's Appearance Moderates How its Decisions are Judged. Int. J. Soc. Robot 2021, 1–10. https://doi.org/10.1007/S12369-020-00738-6.
Landsbergen, D., Coursey, D.H., Loveless, S., Shangraw, R.F., 1997. Decision quality, confidence, and commitment with expert systems: an experimental study. J. Public Admin. Res. Theory 7, 131–157. https://doi.org/10.1093/OXFORDJOURNALS.JPART.A024336.
Langer, M., Landers, R.N., 2021. The future of artificial intelligence at work: a review on effects of decision automation and augmentation on workers targeted by algorithms and third-party observers. Comput. Human Behav. 123, 106878 https://doi.org/10.1016/j.chb.2021.106878.
Lee, M.K., 2018. Understanding perception of algorithmic decisions: fairness, trust, and emotion in response to algorithmic management. Big Data Soc. 5 https://doi.org/10.1177/2053951718756684, 2053951718756684.
Lennartz, S., Dratsch, T., Zopfs, D., Persigehl, T., Maintz, D., Hokamp, N.G., Santos, D.P. dos, 2021. Use and Control of Artificial Intelligence in Patients Across the Medical Workflow: single-Center Questionnaire Study of Patient Perspectives. J. Med. Internet Res. 23, e24221. https://doi.org/10.2196/24221.
Li, Z., Rau, P.L.P., Huang, D., 2020. Who should provide clothing recommendation services: artificial intelligence or human experts? J. Inf. Technol. Res. 13, 113–125. https://doi.org/10.4018/JITR.2020070107.
Liang, H., Xue, Y., 2009. Avoidance of information technology threats: a theoretical perspective. MIS Quarterly 71–90. https://doi.org/10.2307/20650279.

Lim, J.S., O'Connor, M., 1996. Judgmental forecasting with interactive forecasting support systems. Decis. Support Syst. 16, 339–357. https://doi.org/10.1016/0167-9236(95)00009-7.

Litterscheidt, R., Streich, D.J., 2020. Financial education and digital asset management: what's in the black box? J. Behav. Exp. Econ. 87, 101573 https://doi.org/10.1016/J.SOCEC.2020.101573.

Liu, P., Du, Y., Xu, Z., 2019. Machines versus humans: people's biased responses to traffic accidents involving self-driving vehicles. Accident Anal. Prev. 125, 232–240. https://doi.org/10.1016/J.AAP.2019.02.012.

Logg, J.M., Minson, J.A., Moore, D.A., 2019. Algorithm appreciation: people prefer algorithmic to human judgment. Organ. Behav. Hum. Decis. Process. 151, 90–103. https://doi.org/10.1016/j.obhdp.2018.12.005.

Longoni, C., Cian, L., 2020. Artificial Intelligence in Utilitarian vs. Hedonic Contexts: the "Word-of-Machine" Effect. J. Mark. https://doi.org/10.1177/0022242920957347.

Lourenço, C.J.S., Dellaert, B.G.C., Donkers, B., 2020. Whose algorithm says so: the relationships between type of firm, perceptions of trust and expertise, and the acceptance of financial robo-advice. J. Interact. Market. 49, 107–124. https://doi.org/10.1016/J.INTMAR.2019.10.003.

Lu, J., Liang, Y., Duan, H., 2017. Justifying decisions: making choices for others enhances preferences for impoverished options. Soc. Psychol. 48, 92–103. https://doi.org/10.1027/1864-9335/a000302.

Luo, X., Tong, S., Fang, Z., Qu, Z., 2019. Frontiers: machines vs. humans: the impact of artificial intelligence chatbot disclosure on customer purchases. Market. Sci. 38 https://doi.org/10.1287/mksc.2019.1192.

MacKay, J.M., Elam, J.J., 1992. A comparative study of how experts and novices use a decision aid to solve problems in complex knowledge domains. Inf. Syst. Res. 3, 150–172. https://doi.org/10.1287/ISRE.3.2.150.

MacKenzie, I.S., 2012. Human-Computer Interaction: An Empirical Research Perspective. Elsevier, MA 02451, USA.

Madhavan, P., Wiegmann, D.A., 2007. Effects of information source, pedigree, and reliability on operator interaction with decision support systems. Hum. Factors 49, 773–785. https://doi.org/10.1518/001872007X230154.

Madhavan, P., Wiegmann, D.A., Lacson, F.C., 2006. Automation Failures on Tasks Easily Performed by Operators Undermine Trust in Automated Aids. Human factors 48 (2), 241–256. https://doi.org/10.1518/001872006777724408.

Manzey, D., Reichenbach, J., Onnasch, L., 2012. Human Performance Consequences of Automated Decision Aids: the Impact of Degree of Automation and System Experience. J. Cogn. Eng. Decis. Mak. 6, 57–87. https://doi.org/10.1177/1555343411433844.

McBride, M., Carter, L., Ntuen, C., 2012. The impact of personality on nurses' bias towards automated decision aid acceptance. Int. J. Inf. Syst. Change Manag. 6, 132–146. https://doi.org/10.1504/IJISCM.2012.051148.

Mccrae, R.R., Costa, P.T., 1987. Validation of the Five-Factor Model of Personality Across Instruments and Observers. J. Pers. Soc. Psychol.

Merritt, S.M., Ilgen, D.R., 2008. Not all trust is created equal: dispositional and history-based trust in human-automation interactions. Hum. Factors 50, 194–210. https://doi.org/10.1518/001872008X288574.

Meuter, M.L., Ostrom, A.L., Bitner, M.J., Roundtree, R., 2003. The influence of technology anxiety on consumer use and experiences with self-service technologies. J. Bus. Res. 56, 899–906.

Mikalef, P., Gupta, M., 2021. Artificial intelligence capability: conceptualization, measurement calibration, and empirical study on its impact on organizational creativity and firm performance. Inf. Manage. 58, 103434 https://doi.org/10.1016/J.IM.2021.103434.

Mingers, J., Walsham, G., 2010. Toward ethical information systems: The contribution of discourse ethics. MIS Quarterly 34 (4), 833–854. https://doi.org/10.2307/25750707.

Minola, T., Criaco, G., Cassia, L., 2014. Are youth really different? New beliefs for old practices in entrepreneurship. Int. J. Entrepr. Innov. Manage. https://doi.org/10.1504/IJEIM.2014.062881.

Myers, I.B., McCaulley, M.H., Quenk, N.L., Hammer, A.L., 1998. MBTI Manual: A Guide to the Development and Use of the Myers-Briggs Type Indicator, 3rd ed. Consulting Psychologists Press, Palo Alto.

Nass, C., Lee, K.M., 2001. Does computer-synthesized speech manifest personality? Experimental tests of recognition, similarity-attraction, and consistency-attraction. J. Exp. Psychol.: Appl. 7, 171–181. https://doi.org/10.1037/1076-898X.7.3.171.

Newman, D.T., Fast, N.J., Harmon, D.J., 2020. When eliminating bias isn't fair: algorithmic reductionism and procedural justice in human resource decisions. Organ. Behav. Hum. Decis. Process. 160, 149–167. https://doi.org/10.1016/J.OBHDP.2020.03.008.

Niszczota, P., Kaszás, D., 2020. Robo-investment aversion. PLoS One 15, e0239277. https://doi.org/10.1371/journal.pone.0239277.

Noy, N.F., Mcguinness, D.L., 2001. Ontology Development 101: a Guide to Creating Your First Ontology.

Önkal, D., Gönül, M.S., Baets, S.de, 2019. Trusting forecasts. Futures Foresight Sci. 1, e19. https://doi.org/10.1002/FFO2.19.

Önkal, D., Goodwin, P., Thomson, M., Gönül, S., Pollock, A., 2009. The relative influence of advice from human experts and statistical methods on forecast adjustments. J. Behav. Decis. Mak. 22, 390–409. https://doi.org/10.1002/bdm.637.

Orth, U., Robins, R.W., 2014. The Development of Self-Esteem. Curr. Dir. Psychol. Sci. 23, 381–387. https://doi.org/10.1177/0963721414547414.

Pak, R., Fink, N., Price, M., Bass, B., Sturre, L., 2012. Decision support aids with anthropomorphic characteristics influence trust and performance in younger and older adults. Ergonomics 55, 1059–1072. https://doi.org/10.1080/00140139.2012.691554.

Park, J.S., Barber, R., Kirlik, A., Karahalios, K., 2019. A slow algorithm improves users' assessments of the algorithm's accuracy. Proc. ACM Human-Comput. Interact. 3, 15. https://doi.org/10.1145/3359204.

Parker, S.K., Grote, G., 2020. Automation, algorithms, and beyond: why work design matters more than ever in a digital world. Appl. Psychol. An Int. Rev. 2020, 1–45. https://doi.org/10.1111/apps.12241.

Pearson, C.J., Geden, M., Mayhorn, C.B., 2019. Who's the real expert here? Pedigree's unique bias on trust between human and automated advisers. Appl. Ergon. 81, 102907 https://doi.org/10.1016/J.APERGO.2019.102907.

Petticrew, M., Roberts, H., 2008. Systematic reviews in the social sciences: a practical guide.

Prahl, A., Swol, L.van, 2021. Out with the Humans, in with the Machines?: investigating the Behavioral and Psychological Effects of Replacing Human Advisors with a Machine. Human-Mach. Commun. 2, 11. https://doi.org/10.30658/hmc.2.11.

Prahl, A., van Swol, L., 2017. Understanding algorithm aversion: when is advice from automation discounted? J. Forecast. 36, 691–702. https://doi.org/10.1002/for.2464.

Promberger, M., Baron, J., 2006. Do patients trust computers? J. Behav. Decis. 19, 455–468. https://doi.org/10.1002/bdm.542.

Qazi, A., Hussain, F., Wadi, Ali, Al-Fatlawi, A., Raj, R.G., Fayaz, H., Wadi, A., Rahim, N. A., Khan, W.A., 2015. The artificial neural network for solar radiation prediction and designing solar systems: a systematic literature review. J. Clean. Prod. 104, 1–12. https://doi.org/10.1016/j.jclepro.2015.04.041.

Qiu, L., Benbasat, I., 2008. Evaluating anthropomorphic product recommendation agents: a social relationship perspective to designing information systems. J. Manage. Inf. Syst. 25, 145–182. https://doi.org/10.2753/MIS0742-1222250405.

Rapp, A., Curti, L., Boldi, A., 2021. The human side of human-chatbot interaction: a systematic literature review of ten years of research on text-based chatbots 151, 102630.

Rau, P.L.P., Li, Y., Li, D., 2009. Effects of communication style and culture on ability to accept recommendations from robots. Comput. Human Behav. 25, 587–595. https://doi.org/10.1016/J.CHB.2008.12.025.

Sakthivel, V.P., Sathya, P.D., 2021. Fuzzified Coulomb's and Franklin's laws behaved optimization for economic dispatch in multi-area multi-fuel power system. SN Appl. Sci. 3, 1–15. https://doi.org/10.1007/s42452-020-04017-x.

Sanders, N.R., Manrodt, K.B., 2003. The efficacy of using judgmental versus quantitative forecasting methods in practice. Omega (Westport) 31, 511–522. https://doi.org/10.1016/J.OMEGA.2003.08.007.

Shaffer, V.A., Probst, C.A., Merkle, E.C., Arkes, H.R., Medow, M.A., 2012. Why Do Patients Derogate Physicians Who Use a Computer-Based Diagnostic Support System? Med. Decis. Making 33, 108–118. https://doi.org/10.1177/0272989X12453501.

Sharan, N.N., Romano, D.M., 2020. The effects of personality and locus of control on trust in humans versus artificial intelligence. Heliyon 6, e04572. https://doi.org/10.1016/J.HELIYON.2020.E04572.

Sieck, W.R., Arkes, H.R., 2005. The recalcitrance of overconfidence and its contribution to decision aid neglect. J. Behav. Decis. Mak. 18, 29–53. https://doi.org/10.1002/BDM.486.

Slovic, P., Finucane, M.L., Peters, E., MacGregor, D.G., 2004. Risk as analysis and risk as feelings: some thoughts about affect, reason, risk, and rationality. Risk Anal. https://doi.org/10.1111/j.0272-4332.2004.00433.x.

Stein, J.P., Appel, M., Jost, A., Ohler, P., 2020. Matter over mind? How the acceptance of digital entities depends on their appearance, mental prowess, and the interaction between both. Int. J. Hum. Comput. Stud. 142, 102463 https://doi.org/10.1016/J.IJHCS.2020.102463.

Sultana, S., Mozumder, M.H., Ahmed, S.I., 2021. Chasing luck: data-driven prediction, faith, hunch, and cultural norms in rural beting practices. In: Conference on Human Factors in Computing Systems - Proceedings. Association for Computing Machinery. https://doi.org/10.1145/3411764.3445047.

Sutherland, S.C., Harteveld, C., Young, M.E., 2016. Effects of the advisor and environment on requesting and complying with automated advice. ACM Trans. Interact. Intell. Syst. 6, 27. https://doi.org/10.1145/2905370.

Swinney, L., 1999. Consideration of the social context of auditors' reliance on expert system output during evaluation of loan loss reserves. intelligent systems in accounting. Finance Manage. 8, 199–213. https://doi.org/10.1002/(SICI)1099-1174(199909)8:3.

Talukder, M.S., Laato, S., Islam, A.K.M.N., Bao, Y., 2021. Continued use intention of wearable health technologies among the elderly: an enablers and inhibitors perspective. Internet Res. 31, 1611–1640. https://doi.org/10.1108/INTR-10-2020-0586/FULL/HTML.

Taušan, N., Markkula, J., Kuvaja, P., Oivo, M., 2017. Choreography in the embedded systems domain: a systematic literature review. Inf. Softw. Technol. 91, 82–101. https://doi.org/10.1016/J.INFSOF.2017.06.008.

Tandon, A., Dhir, A., Islam, N., Mäntymäki, M., 2020. Blockchain in healthcare: a systematic literature review, synthesizing framework and future research agenda. Comput. Ind. 122, 103290 https://doi.org/10.1016/j.compind.2020.103290.

Teece, D.J., Pisano, G., Shuen, A., 1997. Dynamic Capabilities and Strategic Management. Strategic Manage. J. 18, 509–533. https://doi.org/10.1002/(SICI)1097-0266(199708)18:7<509::AID−SMJ882>3.0.CO;2-Z.

ter Stal, S., Kramer, L.L., Tabak, M., op den Akker, H., Hermens, H., 2020. Design Features of Embodied Conversational Agents in eHealth: a Literature Review. Int. J. Human Comput. Stud. 138, 102409 https://doi.org/10.1016/j.ijhcs.2020.102409.

Thurman, N., Moeller, J., Helberger, N., Trilling, D., 2019. My friends, editors, algorithms, and I: examining audience attitudes to news selection. Digital Journalism 7, 447–469. https://doi.org/10.1080/21670811.2018.1493936.

Toma, I., Delen, D., Moscato, G., 2020. Impact of Loss and Gain Forecasting on the Behavior of Pricing Decision-making. Int. J. Data Sci. Anal. 6, 19. https://doi.org/10.11648/J.IJDSA.20200601.12.
Tomer, G., Mishra, S.K., 2015. Exploring Person Technology Fit and Its Impact on Work Outcomes among IT Professionals. Acad. Manage. Proc. 2015 https://doi.org/10.5465/ambpp.2015.15957abstract, 15957–15957.
Tornatzky, L.G., Fleischer, M., Chakrabarti, A.K., 1990. The Processes of Technological Innovation. Lexington Books, Lexington, Mass.
Tranfield, D., Denyer, D., Smart, P., 2003. Towards a Methodology for Developing Evidence-Informed Management Knowledge by Means of Systematic Review. British J. Manage. https://doi.org/10.1111/1467-8551.00375.
Trocin, C., Mikalef, P., Papamitsiou, Z., Conboy, K., 2021. Responsible AI for Digital Health: a Synthesis and a Research Agenda. Inf. Syst. Front. 2021, 1–19. https://doi.org/10.1007/S10796-021-10146-4.
van Dongen, K., van Maanen, P.P., 2013. A framework for explaining reliance on decision aids. Int. J. Hum. Comput. Stud. 71, 410–424. https://doi.org/10.1016/J.IJHCS.2012.10.018.
van Esch, P., Black, J.S., Arli, D., 2020. Job candidates' reactions to AI-Enabled job application processes. AI Ethics 2 (1), 119–130. https://doi.org/10.1007/S43681-020-00025-0, 2020 1.
Venkatesh, V., Morris, M.G., Davis, G.B., Davis, F.D., 2003. User acceptance of information technology: toward a unified view. MIS Quarterly: Manage. Inf. Syst. 27, 425–478. https://doi.org/10.2307/30036540.
Vimalkumar, M., Sharma, S.K., Singh, J.B., Dwivedi, Y.K., 2021. Okay google, what about my privacy?': user's privacy perceptions and acceptance of voice based digital assistants. Comput. Human Behav. 120, 106763 https://doi.org/10.1016/J.CHB.2021.106763.
Vlačić, B., Corbo, L., Costa e Silva, S., Dabić, M., 2021. The evolving role of artificial intelligence in marketing: a review and research agenda. J. Bus. Res. 128, 187–203. https://doi.org/10.1016/J.JBUSRES.2021.01.055.
Whitecotton, S.M., 1996. The Effects of Experience and a Decision Aid on the Slope, Scatter, and Bias of Earnings Forecasts. Organ. Behav. Hum. Decis. Process. 66, 111–121. https://doi.org/10.1006/OBHD.1996.0042.
Wohlin, C., 2014. Guidelines for Snowballing in Systematic Literature Studies and a Replication in Software Engineering. In: Proceedings of the 18th International Conference on Evaluation and Assessment in Software Engineering. Association for Computing Machinery, pp. 1–10. https://doi.org/10.1145/2601248.2601268.
Worchel, S., Cooper, J., Goethal, G.R., 1988. Understanding Social Psychology. Dorsey Press, Chicago, Ill.
Workman, M., 2005. Expert decision support system use, disuse, and misuse: a study using the theory of planned behavior. Comput. Human Behav. 21, 211–231. https://doi.org/10.1016/J.CHB.2004.03.011.
Yamakawa, Y., Peng, M.W., Deeds, D.L., 2008. What drives new ventures to internationalize from emerging to developed economies? Entrepr.: Theory Practice 32, 59–82. https://doi.org/10.1111/J.1540-6520.2007.00216.X.
Yaniv, I., Kleinberger, E., 2000. Advice taking in decision making: egocentric discounting and reputation formation. Organ. Behav. Hum. Decis. Process. 83, 260–281. https://doi.org/10.1006/obhd.2000.2909.
Yazdani-Asrami, M., Taghipour-Gorjikolaie, M., Song, W., Zhang, M., Yuan, W., 2020. Prediction of nonsinusoidal ac loss of superconducting tapes using artificial intelligence-based models. IEEE Access 8, 207287–207297. https://doi.org/10.1109/ACCESS.2020.3037685.
Yeomans, M., Shah, A., Mullainathan, S., Kleinberg, J., 2019. Making sense of recommendations. J. Behav. Decis. Mak. 32, 403–414. https://doi.org/10.1002/BDM.2118.
Yun, J.H., Lee, E.-J., Kim, D.H., 2021. Behavioral and neural evidence on consumer responses to human doctors and medical artificial intelligence. Psychol. Market. 38, 610–625. https://doi.org/10.1002/MAR.21445.
Yuviler-Gavish, N., Gopher, D., 2011. Effect of descriptive information and experience on automation reliance. Human Factors 53 (3), 230–244. https://doi.org/10.1177/0018720811406725.
Yuviler-Gavish, N., Naseraldin, H., 2016. The effect of previous experience when introducing a decision aid in a decision support system for supply chain management. Cognit. Technol. Work 2 (18), 439–447. https://doi.org/10.1007/S10111-016-0369-Z, 2016 18.
Zahedi, M., Shahin, M., Ali Babar, M., 2016. A systematic review of knowledge sharing challenges and practices in global software development. Int. J. Inf. Manage. 36, 995–1019.
Zhang, D., Xu, J., Li, Haihua, Li, Heming, 2009. A novel image watermarking algorithm with fast processing speed. In: Proceedings - 2009 International Conference on Information Engineering and Computer Science. ICIECS. https://doi.org/10.1109/ICIECS.2009.5362803, 2009.
Zhang, L., Pentina, I., Fan, Y., 2021. Who do you choose? Comparing perceptions of human vs robo-advisor in the context of financial services. J. Serv. Market. https://doi.org/10.1108/JSM-05-2020-0162/FULL/HTML.
Zhou, K.Z., Yim, C.K., Tse, D.K., 2005. The effects of strategic orientations on technology- and market-based breakthrough innovations. J. Mark. 69, 42–60. https://doi.org/10.1509/jmkg.69.2.42.60756.
Zhu, C., Wang, R., 2004. A fast automatic extraction algorithm of elliptic object groups from remote sensing images, in: pattern Recognition Letters. North-Holland, pp. 1471–1478. https://doi.org/10.1016/j.patrec.2004.05.023.

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